



Air Brake and Train Handling Rules

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1.0 General Requirements

1.1 Compliance with FRA Regulations

Inspect and test brake equipment on locomotives and cars according to Federal Railroad Administration (FRA) regulations contained within these rules. This is the responsibility of the employee(s) who perform the work, unless otherwise instructed.

1.2 Qualified Personnel

A Qualified Person or a Qualified Mechanical Inspector may perform inspections and air brake tests.

1.3 Person in Charge of Air Brake Test

The person performing the air brake test is in charge of the train while the test is being conducted. Before permission is given to apply or release the brakes, the person in charge must determine that all crew members are safely positioned.

The employee controlling the locomotive must not apply or release train brakes or move the train without permission from the person performing the air brake test.

1.4 Brake Test General Requirements

Before performing any train air brake test complete the following:

- Properly position all angle cocks and valves.
- Ensure retaining valves are in EX (Direct Exhaust) position.
- Couple the brake pipe air hoses.
- Release hand brakes unless required for securement.
- Ensure the rear car is charged to a minimum of 75-psi.
- If using the Air Flow Method, the Air Flow indicator(s) must be calibrated.

1.5 Handheld Gauges

Handheld air gauges need to be verified as accurate to within +/- 3-psi. Accuracy can be verified by connecting the handheld air gauge to the brake pipe air hose of a locomotive and comparing the pressure against the locomotive brake pipe gauge. If the pressure displayed on the handheld gauge exceeds tolerance, it must not be used. Handheld gauges must be tested annually for accuracy.

2.0 Air Hoses

2.1 Coupling and Securement

When coupling air hoses employees must:

- Prior to coupling, check for and remove foreign matter and blow condensation from the brake pipe hose.
- Verify that the brake pipe glad hands are at least 5 inches above the top of the rail.
- When equipped, ensure air hose supports are connected.

2.2 Use of Adapter Hoses

The following requirements will apply to the use of adapter hoses:

- Adapter hoses must not be used when departing an initial terminal unless authorized by Mechanical.
- Where required enroute, only use an adapter hose to move equipment to the next terminal where inspections and repairs can be made.

3.0 Charging the Brake System

3.1 Air Brake System Charge Requirements

The air brake system must be charged to the pressure(s) required by this section.

A handheld gauge or EOT located at the rear of a train may be used to determine that the air brake system has been charged to a specified pressure and brake pipe pressure has been restored. A distributed power consist located at the rear of a train may be used to determine that flow is within specified limits and brake pipe pressure has been restored.

Do not charge a train's air brake system with more than one automatic brake valve cut in unless utilizing distributed power. If unable to maintain a minimum main reservoir pressure of 105-psi, the throttle may be advanced to notch 4 with the generator field switch open. This does not apply to locomotives equipped with an electronic air compressor.

4.0 Train Brake Pipe Leakage Tests

4.1 Leakage Test Required

Brake pipe leakage must be tested when conducting Class I or Class IA air brake tests. The Air Flow Method is the preferred method to test brake pipe leakage.

4.2 Air Flow Method (AFM)

To utilize the AFM, a train must be equipped with the following:

- A 26-L or equivalent pressure maintaining automatic brake valve.
- Operative Air Flow Indicator with a direct reading of air flow in increments no greater than 10 cubic feet per minute (CFM).
- Operational HOT and EOT or handheld gauge.

Perform the AFM test as follows:

- Charge the air brake system to 75-psi.
- Using the AFM indicator, ensure air flow does not exceed:
 - 60 CFM for conventional trains.
 - 90 CFM combined air flow with no more than 60 CFM at any consist for distributed power trains.
- If CFM exceeds limits above, inspect the train for leakage, make necessary repairs, and retest.

4.3 Brake Pipe Leakage Test (Non-AFM)

If an AFM test cannot be performed, conduct the following test:

- Ensure the brake system is charged to at least 75-psi.
- Wait for the signal to apply the brakes.
- When signal is received, make a 20-psi reduction with the automatic brake.
- Allow the brake pipe exhaust to stop.
- Wait 60 seconds.
- Cut out the automatic brake.
- Wait an additional 60 seconds.
- Do not actuate the independent brake or move the automatic brake handle until test is complete.
- Time brake pipe leakage for 60 seconds.
 - If brake pipe leakage does not exceed 5-psi, test is complete.
 - If brake pipe leakage exceeds 5-psi, inspect train for leakage, make necessary repairs, and retest.
- When a signal to release the brakes is received, move the automatic brake handle to the release position and cut in the automatic brake valve.

5.0 Brake Test Overview

5.1 Brake Test Overview Chart

Test	Location	Performed By	Minimum Air Requirements	Car Inspection Per Section 2	Leakage Test Required	Reduction Required (psi)	Operative Brakes	Cars Inspected
Class I	Initial Terminal/ Enroute	QMI or QP	75-psi	X	X	20	Initial Terminal 100% Enroute - See Operative Brakes	All
Class IA	Enroute	QMI or QP	75-psi	X	X	20	100% See Operative Brakes	All
Class III	Enroute	QP	75-psi			20	Last piece of Equipment - can be verified with EOT.	-
Transfer	Initial Terminal/ Enroute	QMI or QP	75-psi	X		20	100% must apply. See Operative Brakes	All

6.0 Air Brake Status

6.1 Operative Brakes

Air brakes must be operative on 100 percent of cars at the initial terminal. Cars qualified by a Class IA or transfer test and cars picked up enroute, must have 100 percent operative brakes unless properly tagged and approved by a Qualified Mechanical Inspector.

At least 85 percent of the cars in a train must have operative brakes.

6.2 Inoperative Brakes

Air brakes on all cars must be operative, except:

- Cars with air brakes that fail enroute may be handled to the next forward repair location.
- When properly tagged by a Qualified Mechanical Inspector, cars may be moved for repair under the following air brake tests:
 - Class I, at locations other than the initial terminal.
 - Class IA.
 - Transfer.

No more than two consecutive cars in a train may have inoperative brakes.

The rear three cars of a train must have operative brakes. If the rear car's air brakes become inoperative enroute, movement may continue under the direction of Mechanical as follows:

- Test the hand brake on the disabled car.
 - If the hand brake is inoperative, do not move the car until repaired.
- Chain, strap, or cable the disabled car to the adjacent car.
- Ensure the brake pipe is connected and open between the disabled car and train.
 - If necessary to close the angle cock on the disabled car, ensure the angle cock on the adjacent car remains open.
 - This will ensure that an emergency brake application can propagate throughout the train in the event the disabled car becomes separated.
- Move the car directly to the first available location where it can be switched between cars with operative brakes or set out of the train.

Cars with inoperative brakes must be reported to Mechanical and recorded on the Crew Information Form.

7.0 Class I Air Brake Test

7.1 Class I Air Brake Test Requirements

Conduct a Class I air brake test on a train:

- Where the train is originally assembled.
- When the train has been off air for more than 24 hours.
- When a train is received in interchange or a location where the train consist is changed other than by:
 - Changing locomotive consist.
 - Adding a previously tested car or solid block of cars.
 - Removing a car or solid block of cars.
 - Removing defective equipment.
 - Any combination of the changes listed above.
- A unit or cycle train has traveled 3,000 miles since its last Class I.

Conduct a Class I air brake test on a portion of a train:

- When car(s) have been off air for more than 24 hours.
- When one or more cars are added to a train that have not been pretested by a Class I air brake test.
- When a solid block of cars are added to the train that are comprised of cars from more than one previous train.
- When a car or solid block of cars from a single previous train have not remained continuously and consecutively coupled with the train line remaining connected, except:
 - When cars are separated into multiple solid blocks due to space or trackage constraints.
 - Cars must be recoupled in the same order as removed.

7.2 Class I Air Brake Test Procedure

To perform a Class I air brake test, complete the following:

- Charge the air brake system to at least 75-psi.
- Test brake pipe leakage.
- When signal is received, make a 20-psi reduction with the automatic brake.
 - Brakes must remain applied for a minimum of three minutes and until a release initiated.
 - If test is unsuccessful, retest cars until compliant or set out of train.
 - Brakes must be operative on 100 percent of cars.
- Verify brakes apply on each car, brake rigging does not bind or foul, and piston travel is within applicable limits.
- Inspect each side of each car to examine and observe the functioning of all moving parts of the brake system and comply with the freight car inspection requirements.
 - Equipment must be stationary during the inspection.
- When signal is received, move the automatic brake handle to release.
- Inspect each car and verify that brakes are released.
 - Release may be verified during a roll-by inspection, not exceeding 10 mph.

7.3 Engineer Notification

The locomotive engineer must be notified with the results of a Class I air brake test. This information must be recorded on the Crew Information Form or electronic record and retained in the controlling locomotive cab until the train reaches its destination.

At minimum, notification will include:

- Test completion status.
- Date and time.
- Location of test.
- Number of cars inspected.
- Name of person(s) performing test.

8.0 Class IA Air Brake Test – 1,000 Mile Inspection

8.1 Class IA Air Brake Test Procedure

To perform a Class IA air brake test, complete the following:

- Charge the air brake system to at least 75-psi.
- Test brake pipe leakage.
- When signal is received, make a 20-psi reduction with the automatic brake.
 - Brakes must remain applied for a minimum of three minutes and until a release initiated.
 - If test is unsuccessful, retest cars until compliant or set out of train.
 - Brakes must be operative on 100 percent of cars except as outlined by Operative Brakes of this section.
- Verify brakes apply on each car, brake rigging does not bind or foul, and piston travel is within applicable limits.
- Inspect each side of each car to examine and observe the functioning of all moving parts of the brake system and comply with the freight car inspection requirements.
 - Equipment must be stationary during the inspection.

9.0 Class III Air Brake Test

9.1 Class III Air Brake Test Requirements

Conduct a Class III air brake test on a train:

- When a locomotive consist is changed.
- Adding or removing a car or solid block of cars.
- Adding locomotive(s) to a train after receiving a brake test from a yard air source.
- Adding or removing distributed power or helper locomotive(s) from anywhere in a train.
- When a caboose is added or removed.

9.2 Class III Air Brake Test Procedures

To perform a Class III air brake test, complete the following:

- Charge the air brake system to at least 75-psi.
- Make a 20-psi reduction with the automatic brake.
- Distributed power trains must follow Train Check procedures.
- Verify the brakes apply on the rear car.
 - A 5-psi decrease on EOT or rear of train distributed power locomotive can be used to determine the brakes are applied.
- Move the automatic brake handle to release.
- Verify the brakes release on the rear car.
 - A 5-psi increase on EOT or rear of train distributed power locomotive can be used to determine the brakes are released.

When an EOT or rear of train distributed power locomotive cannot be used, the rear car must be visually inspected to determine application and release of the brakes.

10.0 Transfer Train Air Brake Test

10.1 Transfer Air Brake Test Requirements

A transfer test may be used on a train movement that does not exceed 20 miles in one direction.

10.2 Transfer Air Brake Test Procedures

To perform a Transfer Train Air Brake Test, complete the following:

- Charge the air brake system to at least 75-psi.
- When signal is received, make a 20-psi reduction with the automatic brake.
- Verify the brakes apply on each car and remain applied until a release is initiated.
 - Brakes must apply on each car, except as outlined by Operative Brakes in this section.

11.0 Extended Haul Trains

11.1 Extended Haul Train Requirements

Trains designated as Extended Haul may be operated greater than 1,000 miles but cannot exceed 1,500 miles before a subsequent Class I or Class IA air brake test is performed.

The following requirements apply to Extended Haul Trains:

- Air brake test and inspection must be performed by a Qualified Mechanical Inspector.
- The train may not depart an initial terminal with mechanical defects or inoperative brakes.
- The train may not make more than one pick up and more than one set out between the initial terminal and intermediate inspection point.
 - This does not apply to defective equipment set out enroute.
- Any car or solid block of cars added enroute must be pretested by a Qualified Mechanical Inspector.

12.0 Running Air Brake Test

12.1 Passenger Trains

Perform a Running air brake test on passenger trains and trains consisting entirely of business cars when:

- The train leaves the initial terminal.
- Locomotive, engine crew, train crew, or operating ends have been changed.
- Any angle cocks or cutout cocks have been opened or closed.
- An initial terminal air brake test has been conducted.
- The train reaches points designated by the timetable or general order.
- The train has struck debris on the track.

12.2 All Trains

Conduct a running air brake test when:

- Inclement weather conditions are present which may cause snow or ice build-up to occur between the brake shoes and wheels.
- Snow is above the top of the rail or where icing conditions may exist and train is approaching:
 - A meeting, passing, or stopping point.
 - A signal indication which will require the train to stop.

The following requirements apply to running air brake tests:

- The locomotive engineer must make a brake pipe reduction sufficiently in advance of prescribed locations to determine that the brakes are working properly.
- If braking effort is not sufficient, the train must be stopped by a full service brake application using full dynamic brakes, consistent with proper train handling.
- If deemed necessary, the locomotive engineer must make an emergency brake application without hesitation.
- After stop is made, the train must be inspected to determine if brake rigging and shoes are free of snow and ice before proceeding.
- Immediately after proceeding, perform a running brake test at a safe speed to determine if the brakes respond as intended.

12.3 Running Air Brake Test Procedures

To perform a Running Air Brake Test, complete the following:

- When speed reaches a level adequate to ensure continued motion, make an automatic brake reduction sufficient to create an evident retarding force.
- Validate that train brakes are responding as intended.
- If brakes are operating properly, release the automatic brake and proceed.

If the Running Air Brake Test fails, complete the following:

- Immediately stop the train.
- Make a walking inspection of the train to identify issues with the braking system.
- When issues are identified, correct the problem(s) and notify the train dispatcher.
- Before proceeding, perform a Class III air brake test.
- After proceeding, perform another running brake test.

13.0 Enroute Brake Pipe Pressure Requirements

An enroute train must be stopped and inspected for leaks at the next available location when air flow or gradient do not return to the following parameters within a reasonable amount of time. The following requirements apply:

- Conventional trains:
 - Brake pipe gradient must not exceed 15-psi.
 - When qualified by an AFM test, air flow must not exceed 60 CFM.
- Distributed Power trains:
 - Combined total air flow must not exceed 90 CFM or 60 CFM at any locomotive consist.

In the event a brake application cannot be propagated or an abnormal change or loss in brake pipe pressure occurs with the automatic brake released, the train must be immediately stopped and inspected.

14.0 Overcharged Brake Conditions

An overcharged brake system occurs when brake equipment on cars or locomotives is charged to a higher pressure than the regulating valve setting.

To eliminate an overcharge on a train:

- Make a full service reduction with the automatic brake.
- Wait two minutes.
- Move the automatic brake handle to release.

15.0 Block Swaps

A Class I Air test is not required to be performed on a solid block of cars intended for block swap when the solid block of cars is comprised of cars from a single previous train and have:

- Previously received a Class I brake test.
- Remained continuously and consecutively coupled together with the train line connected since being removed from the previous train.
- Not been off air for more than 24 hours.

Additional requirements:

- A freight car inspection must be performed at the location where the car(s) are added to the train.
- The solid block of cars from a previous train must have the required train brake status information recorded on a Crew Information Form.

16.0 Yard Air

16.1 Yard Air Requirements

The following instructions govern the use of yard air systems. Employees are prohibited from using yard air sources until proper training has been completed and a clear understanding of respective components has been demonstrated.

16.2 Connecting Yard Air

To connect yard air:

- Ensure equipment is secured.
- Uncouple from the equipment to be left standing.
- To ensure the yard air hose remains visible to employees, route the yard air hose through the side of the car being connected using a handhold or other location.
- Couple the air hose connection between the cars and yard air system.
- Ensure the angle cock on the equipment coupled to the air source is open.
- Open the valve on the yard air source.

16.3 Disconnecting Yard Air

To disconnect yard air:

- Prior to coupling to equipment, ensure yard air hoses are not positioned in a manner that will cause them to be damaged.
- Verify equipment is properly secured and couple to equipment.
- Close the angle cock on the equipment that is coupled to the yard air line.
- Close the air supply valve from the yard air line.
- Release the pressure from the hose connecting the yard air source to the equipment.
- Once pressure has been exhausted, remove the air hose from the cars.
- Leave the yard air line in a manner that will not create a tripping hazard or be damaged by equipment.
- Couple the brake pipe between equipment.

- Once the locomotive engineer acknowledges the brake system is charged and it is safe to do so, release the hand brakes.
- Perform a Class III air brake test.

17.0 Interchange Requirements

17.1 Trains Entering the United States of America (USA) from Canada

Trains which are destined for the USA must have 100 percent operative brakes from the initial terminal or an enroute safety inspection location. No train will be permitted to operate into the USA with less than 85 percent operative brakes. Cars destined for the USA with inoperative brakes or any other safety defect must not be moved beyond the first location where repairs can reasonably be made.

17.2 Trains Entering Canada from the USA

Trains may enter Canada when the Crew Information Form indicates a Class I or Class IA air brake test has been performed.

17.3 Trains Received from Foreign Carrier

Trains received from a foreign carrier must have the train brake status information transferred to the Crew Information Form. The air brake information received from a foreign carrier will be kept with the Crew Information Form. In the event train brake status information is unavailable, contact the train dispatcher.

18.0 Crew Information Form and Train Brake Status

18.1 Crew Information Form

The Crew Information Form consists of the following sections:

- EOT Test information:
 - Contains a record of train identification, lead locomotive, and EOT test information.
- Defects:
 - Portion of form to record locomotive defects.
- Dynamic Brake Status Chart:
 - Contains dynamic brake status information.
- Hot Box Detector Information.
- Results of Inbound Train Inspection.
- Emergency Brake Application Inspection Report.
- Train Brake Status:
 - Contains information related to initial air brake tests and work performed enroute.

18.2 Recording Train Brake Status

The results of a Class I air brake test, including pretested blocks of cars, must be recorded in the Train Brake Status Section of the Crew Information Form.

At locations where a train receives a Class I or Class IA air brake test, the conductor must record the following information on the Crew Information Form:

- Date.
- Name of conductor or Mechanical employee performing test.
- Train Identification.
- Location.
- Number of cars on the train.
- Air brake status of cars in train:
 - “All OK” when no air brakes are cut out.
 - Specific car numbers when cars are cut out.

Class I No 1, No 1A, Class 1, Class 1A	brake test performed at	St. Paul location	by	Stephen conductor/carman
at 0600 time	2/20/2022 date	Number of cars tested	120	All “OK” ☒ or “cut-out” as listed below

18.3 Updating Train Brake Status

The train brake status must be updated each time cars are pick up or set out of the train. Prior to arrival at a crew change point, the conductor must update the Train Brake Status section of the Crew Information Form. The following information must be recorded:

- Date.
- Name of conductor.
- Total number of cars in the train.
- Number of cars picked up or set out.
- Type of air brake test performed.
- Air brake status of cars in train:
 - “All OK” when no air brakes are cut out.
 - Specific car numbers when cars are cut out.

18.4 Crew Information Form Handling

The conductor is responsible for handling of the Crew Information Form. The original and subsequent Crew Information Forms must be kept with other train documents while enroute and until destination.

Example:

Date	Conductor	Station Name	Car Total	Equipment Lifted / Setoff enroute "Brake Test Results" Number of cars cut-out or all "OK" (Run Through)	Rear Car Pressure*
2/20	Van Horne	River Junction	140	Pick up 20 cars – Class I Air Test – All OK	89
2/20	Van Horne	Marquette	140	All OK	89
2/20	Stilwell	Dubuque	115	Set Out 25 cars – All OK	89
2/20	Stilwell	Nahant	115	All OK	89
2/20	Belle	Muscatine	145	Pick up 30 cars – Class I Air Test – All OK	89
2/20	Belle	Ottumwa	145	All OK	90
2/21	Crump	Cotter	145	1 car cut out (record car number)	90
2/21	Crump	Kansas City	105	Set out 30 cars – All OK	90
2/21	Bowen	Blue Valley	140	Pick up 35 cars – Class I Air Test– All OK	90
2/21	Bowen	Pittsburg	140	All OK	90

Section 2 – Car Equipment, Inspections, and Components

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1.0 Car Inspection

1.1 Car Inspection Requirements

At each location where a freight car is placed in a train and mechanical forces are not on duty for the purpose of inspecting freight cars, the freight car shall, as a minimum, be inspected for the following conditions:

- Car body:
 - Leaning.
 - Sagging downward.
 - Improperly positioned truck.
 - Object dragging below.
 - Object extending from side.
 - Door insecurely attached.
 - Broken or missing safety appliance.
- Lading leaking from a placarded hazardous material car.
- Insecure coupling device.
- Overheated wheel or journal.
- Broken or cracked wheel.
- Brake that fails to release.
- Any other apparent safety hazard likely to cause an accident or casualty before the train arrives at its destination.

This inspection must be performed on each side of all cars.

2.0 Freight Car End, Platform, and Wheel Identification

2.1 Car End Identification

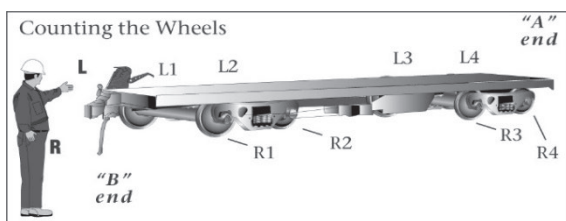
Identify car ends as follows:

- On cars with one hand brake, the “B” end of the car is the end with the hand brake and the other end is the “A” end.
- On cars with more than one hand brake, the letters “A” and “B” are stenciled on the appropriate ends of the car.

2.2 Wheel Identification

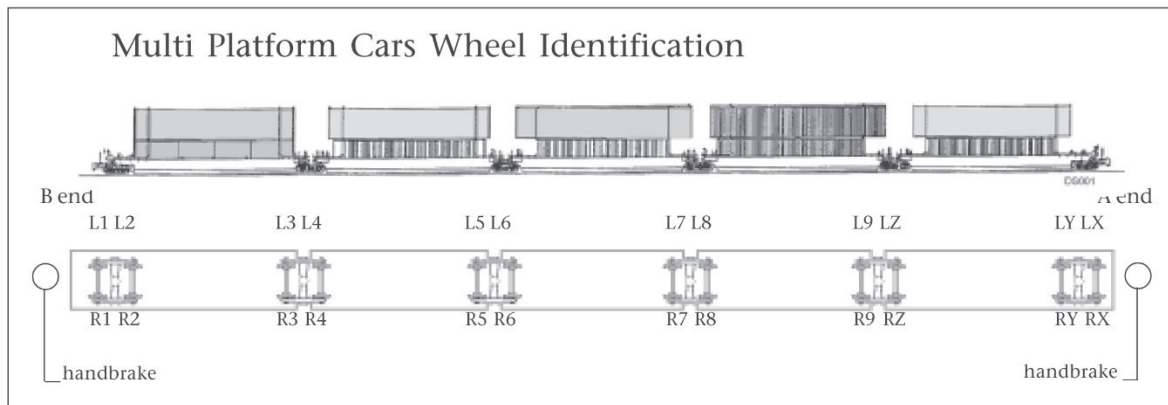
Car wheel numbers are identified as follows:

- Axles are designated from the “B” end of the car with number 1 being the axle closest to the “B” end.
- Wheels and journals are designated as either left or right as viewed from the “B” end.
- Specific wheels are identified using both the axle and wheel designation.



Identify the wheels of a multi-platform car as follows:

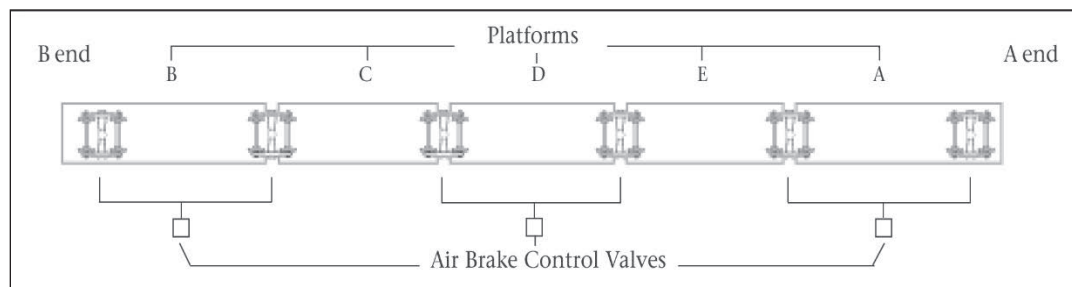
- Axles are designated from the “B” end of the car with the number 1 axle closest to the “B” end.
- Wheels and journals are designated as either left or right as viewed from the “B” end.
- Specific wheels are identified using the axle and wheel designation.
- Wheels will be designated using number 1 through 9 and then alphabet characters from “Z” to “A” for each subsequent axle.



2.3 Platform Identification

Identify platforms as follows:

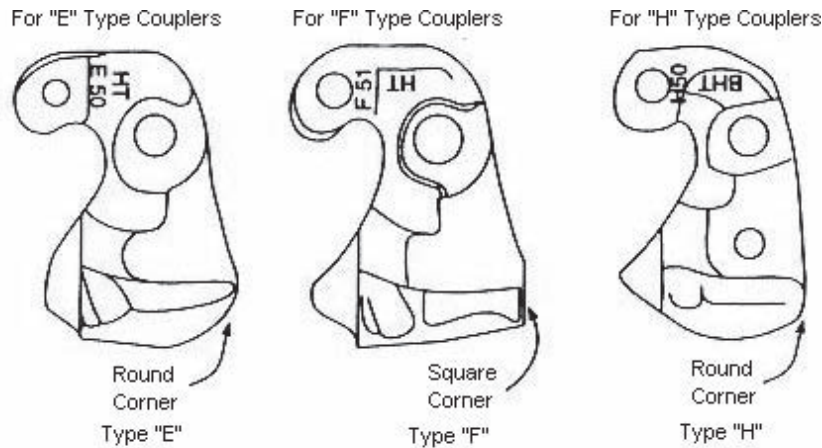
- Multi-platform cars have stencils indicating “A” and “B” ends of the equipment.
- The “B” platform is located on the “B” end of the car, while the “A” platform is located on the “A” end.
- The remainder of the platforms are identified as “C” through “E” beginning at the platform immediately adjacent to the “B” end.



3.0 Coupler Identification

3.1 Coupler Assembly Types

There are three main types of coupler assemblies used in North America. Each coupler head and knuckle is marked with a letter indicating its type, E, F, or H. Knuckles are not interchangeable.



3.2 Assembly type descriptions

- Type E – Most common assembly used on all types of freight cars.
- Type F – Interlocking coupler assembly used on most intermodal and some bulk commodity equipment.
- Type H – “Tight Lock” coupler assembly used on passenger equipment and some work equipment.

4.0 Retaining Valves

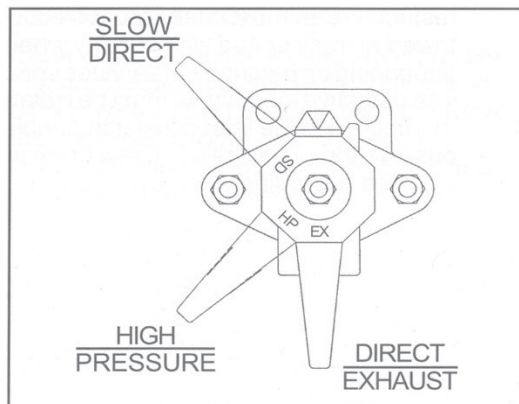
4.1 Retaining Valve Location

The retaining valve controls brake cylinder pressure exhaust on a freight car. All freight cars have retaining valves located at the “B” end of the car or on the side near the control valve.

4.2 Three-Position Retaining Valve

The three-position retaining valve consists of the following positions:

- Exhaust (EX) – Allows unrestricted passage of brake cylinder pressure to exhaust to atmosphere; Handle is turned down.
- High Pressure (HP) – Restrict the exhaust of brake cylinder pressure at a controlled rate and retains 20 pounds of brake cylinder pressure; Handle is 45 degrees below horizontal.
- Slow Direct (SD): Restrict the exhaust of brake cylinder pressure at a controlled rate to provide a blow down time of approximately 86 seconds and continues to exhaust until all brake cylinder pressure is vented to atmosphere; Handle is 45 degrees above horizontal.



4.3 Four-Position Retaining Valve

The four-position retaining valve includes the positions listed above and one additional position:

- Low Pressure (LP): Exhausts brake cylinder pressure to 10-psi; Handle is horizontal.

4.4 Retaining Valve Use

The normal position for a retaining valve is Direct Exhaust. When retaining valves are required, the High-Pressure setting must be used and movement must not exceed 20 mph.

5.0 Piston Travel

5.1 Piston Travel Requirements

Follow the piston travel requirements outlined on the car's stencil, sticker, or badge plate. When required, this information will be displayed on each side of the car body in a location near the brake cylinder. If no stencil, sticker, or badge plate is available, piston travel must be within the following guidelines.

5.2 Body-Mounted Brake Cylinders

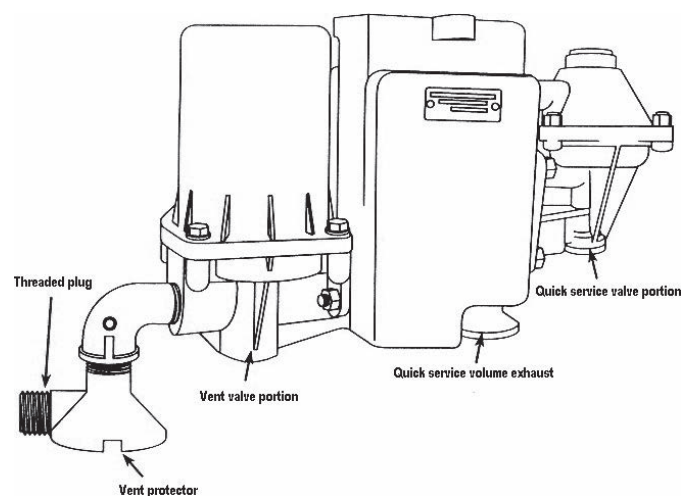
- Initial Terminal
 - Piston travel on cars equipped with 8 ½ or 10-inch diameter brake cylinders must be 6 to 9 inches; piston travel outside of this range must be adjusted to 7 ½ inches.
 - Cars not equipped with 8 ½ or 10-inch diameter brake cylinders must be within the limits indicated on stencil, sticker, or badge plate.
- Intermediate locations
 - Piston travel on cars equipped with 8 ½ or 10-inch brake cylinders must be 6 to 10 ½ inches.
 - Piston travel on cars not equipped with 8 ½ or 10-inch diameter brake cylinders must be within the limits indicated on stencil, sticker, or badge plate. For maximum travel, use the "Piston Travel Ineffective in excess of" column on brake travel stencil.

5.3 Truck-Mounted Brake Cylinders

Piston travel must be within the limits of the travel indicator when brakes are set and provide brake shoe clearance when brakes are released.

6.0 Vent Valve

The number 8 vent valve (on the control valve) transmits emergency brake pipe reductions. If the vent valve fails to reset after an emergency brake application, causing a continuous blow at the exhaust port, plug the valve by removing the vent protector and screwing in the threaded plug.



7.0 Cutting Out Air Brake Equipment

Car air brakes may only be cut out in the when brake equipment is defective. To cut out control valves, close the branch pipe cut out cock and drain the air reservoirs completely by operating the brake cylinder release rod.

8.0 Bleeding Rail Cars

Cars may only be bled off when making repairs to the brake system, cutting out defect cars, or switching.

9.0 Flat Wheels and Shell Spots – Freight Cars

While inspecting freight car wheels, measure and report as follows:

- Determine the length of the flat or shelled area.
- Notify the Dispatcher and Mechanical when the flat spot or shelled area exceeds 1 inch.
- Use the table below to determine if equipment restrictions apply.

Maximum Speed Permitted for Flat Wheel and Shell Spots - Freight Cars		
Length of Single Flat or Shell Spot	Length of Two Adjoining Flat or Shell Spots Less Than 1½" Apart	Maximum Speed
Less than 2¼"	1½" or less	Maximum authorized speed.
2¼" to 2½"	Over 1½" to 2"	50 mph
2½" or more	2" or more	10 mph. Set out at first available point but not before reporting to train dispatcher.

10.0 Towing Equipment

10.1 General Requirements

The following will apply to the use of tow cables:

- Equipment may only be towed under the direction of Mechanical.
- Mechanical will be responsible to apply tow cables and other required components.
- When uncertainty exists regarding the safe movement of a car with a tow cable, the equipment must be moved with a locomotive.
- Do not stand near a tow cable while in use.
- Tow cables must not be connected to axles or truck side frames.
- Except for multi-platform cars, do not attempt to tow more than one car at a time.
- When possible, apply a light hand brake to the car being towed to avoid slack action and shock loading of the tow cable.
- While towing, an employee must be in position to operate the hand brake of the car being towed, unless doing so would place the employee in the vicinity of the tow cable.

10.2 Tow Cable Weight Limits

The following grade limitations will apply when using a 7/8-inch tow cable:

Max Car Weight by Grade

- Level to 0.7 percent grade: 400 tons.
- 0.7 to 1.0 percent grade: 330 tons.
- 1.0 to 1.5 percent grade: 260 tons.
- 1.5 to 2.0 percent grade: 210 tons.
- 2.0 to 2.5 percent grade: 180 tons.
- 2.5 to 3.0 percent or greater: 155 tons.
- 3.0 percent or greater: Not permitted.

11.0 Reporting and Recording Defects

11.1 Reporting Defects

Defects must be reported to Mechanical and recorded on the proper form or in the required system.

11.2 Recording Defects

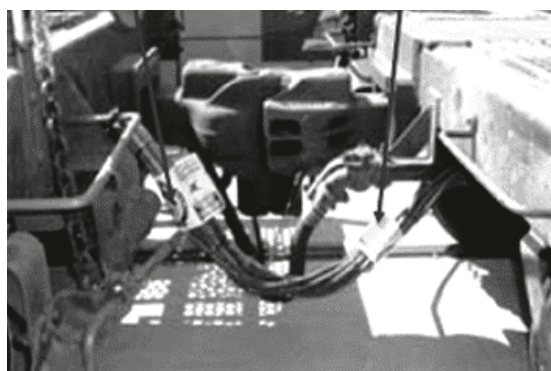
The following details must be recorded on the Crew Information Form:

- Trackside Warning Detector alarms and details of inspections.
- Emergency brake applications.
- Train inspection results.
- Location and number of hand brakes applied.
- Details of equipment with inoperative air brakes.
- Details of defective equipment being moved in the train.

12.0 Container Rail Cars with Electric Power Cables

12.1 Warning Tags

When electrical cables are connected between cars that cannot be uncoupled, the following warning tags will be applied to the electrical cable(s) between cars:



12.2 Consist Warning Message – Bad Order Set Out

The train consist will display a message notifying crews when applicable equipment cannot be uncoupled account electrical supply cable connections between cars.

Example message:

```
*****  
WARNING: DO NOT UNCOUPLE FROM CAR  
DTTX 1, DTTX 2, DTTX 3  
ACCOUNT ELECTRIC CABLE CONNECTIONS  
*****
```

Cars connected by electrical cables cannot be set out individually. If a defect occurs enroute, all electrically connected cars must be set out together.

13.0 Equipment Requiring Supplemental Air Source

Snowplows, spreaders, air dumps or other work equipment requiring a supplementary source of air, must be connected to the locomotive's main reservoir hose.

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1.0 General Requirements

Locomotive engineers must ensure:

- Each locomotive in service is inspected at least once during each calendar day.
- Car body compartment, engine room, and end doors of locomotive(s) are closed when not in use.
- On unoccupied locomotives:
 - Awnings are down.
 - Mirrors are pulled in.
 - Windows and doors are closed.
- High voltage cabinets remain closed.

When locomotive(s) are in motion, personnel may only move from one locomotive to an adjoining locomotive when:

- A job briefing has been performed concerning:
 - Movement between locomotives.
 - Required speed to ensure safe passage.
- Both locomotives are equipped with walkways.
- Safety chains are connected and provide a continuous barrier.
- If there is doubt or uncertainty regarding safe passage, movement between locomotives is prohibited.

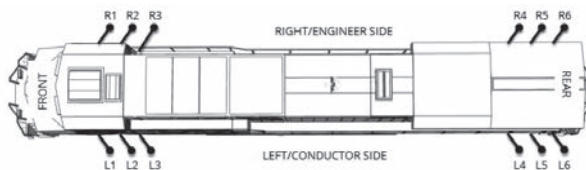
Employees are restricted from opening engine covers and electrical cabinet doors unless authorized or required in the case of emergency.

In case of fire to locomotives or rail diesel cars, engines and fuel pumps should be stopped, emergency fuel cutoffs tripped, and battery switches opened as soon as possible.

Locomotives and rail diesel cars must not be stopped over flame type switch heaters. If unavoidable, switch heaters must be turned off by the dispatcher or equipment promptly moved.

2.0 Locomotive Orientation and Wheel Numbering

The front of the locomotive is designated by the letter “F” stenciled on each side of the locomotive. Wheels are numbered consecutively as illustrated in the following diagram.



3.0 Standard Air Pressures

The following standards air pressures apply:

- With the automatic brake cut in and in the release position, the main reservoir pressure must be at least 15-psi higher than the standard locomotive brake pipe pressure.
- The standard equalizing reservoir pressure setting is 90-psi.
- With independent brake handle fully applied, the brake cylinder pressure must be within 3-psi of the pressure posted in the locomotive cab.
 - If the brake cylinder pressure is unknown, contact Mechanical for instructions.

4.0 Taking Charge of Locomotive(s)

When taking charge of locomotive(s), the locomotive engineer must:

- Determine the locomotive daily inspection status and perform a locomotive daily inspection, if required.
- Perform a locomotive air brake test, if required.
- Evaluate the operating condition of locomotive(s).
- Confirm that total driving axle and dynamic brake factors do not exceed the permissible limits and record information on the Crew Information Form.
- Ensure that the locomotive air brake system is properly conditioned.
- Ensure that the headlights, ditch lights, bell, and whistle are working on the controlling locomotive.
- Ensure the locomotive brakes and devices for regulating air pressures, including the automatic and independent brake valves, are operating as intended.
- Check locomotive fuel level(s) and report to the proper authority when fuel levels are below:
 - 1,000 gallons on road locomotives.
 - 300 gallons on yard locomotives.
- Record locomotive defects on the Crew Information Form and report to Mechanical.

An inbound locomotive engineer must conduct a job briefing with the outbound locomotive engineer regarding the condition of locomotive(s). If a job briefing cannot be conducted, the condition of locomotive(s) must be recorded on the Crew Information Form.

5.0 Locomotive Daily Inspection

5.1 Inspection Requirement

Locomotive engineers are responsible for determining if locomotive(s) need to be inspected by reviewing the Locomotive Daily Inspection Report in each locomotive cab. The card will indicate the date and time of the last inspection. Locomotive engineers must ensure that each locomotive in their charge, including locomotives picked up enroute, are inspected each calendar day the locomotive is in service. A calendar day is comprised of the time between 00:01 and 23:59 hours. The Locomotive Daily Inspection Report on the controlling locomotive may be used to determine the inspection date for all locomotives in the train.

5.2 Inspected Previous Calendar Day

If the Locomotive Daily Inspection Report indicates that the locomotive was inspected the previous calendar day, then complete the current daily inspection before the end of the tour of duty. Contact the train dispatcher, yardmaster, or other proper authority to determine where to complete the daily inspection. The locomotive engineer may be relieved from requirements to perform a daily inspection when informed by the proper authority that Mechanical or the relieving locomotive engineer will inspect the locomotive(s).

When possible, perform locomotive daily inspections during daylight hours or at the beginning of a shift.

5.3 Not Inspected Previous Calendar Day

If the Locomotive Daily Inspection Report indicates that the locomotive was not inspected during the previous day, or if there is no record on the locomotive, inspect the locomotive before it is placed in service on the current day. When not used, the words “Not Used” must be written on each applicable date line of the Locomotive Daily Inspection Report.

5.4 Locomotives Picked Up or Set Out Enroute

Locomotives picked up enroute must not have an inspection date that precedes the inspection date of the controlling locomotive. Locomotives set out enroute must have a current calendar day inspection. The locomotive engineer may be relieved from requirements to perform a daily inspection when instructed by the proper authority.

5.5 Inspection Items

Unless functioning properly, the following inspection items will be considered non-complying defects.

5.5.1 Locomotive Cab

From the locomotive cab, ensure:

- FRA Form F 6180-49A (blue card) is displayed and up to date.
- Cab seats are securely mounted and braced.
- Cab doors are equipped with a device to secure the door.
- Cab windows of the lead locomotive provide an undistorted view of the right-of-way for the crew from their normal position in the cab.
- Cab floors, passageways, and compartments are free from oil, water or any obstruction that creates a slipping, tripping or fire hazard, and floors have been properly treated to provide secure footing.
- The cab of the lead locomotive provides proper ventilation with a heating arrangement that maintains a temperature of at least 60 degrees Fahrenheit 6 inches above the center of each seat in the cab.
- Each air gauge registers within 3-psi of the required pressure.
- A controlling locomotive operating at speeds greater than 20 mph has an operative speed indicator.
- Locomotive is equipped with a fusee container.
- Cab lights provide sufficient illumination for control instruments and gauges.
- At least one head light is operational on each end of the locomotive consist.
- Both ditch lights on the leading end of the controlling locomotive are operational.

- All traction motors are cut in.
 - This does not apply to GE-8 DC, GE-9DC, and all AC locomotives.
- The dynamic brake, if equipped, has not been defective for more than 30 days.
- Horn and bell operate on the lead locomotive.
- The locomotive toilet facility is sanitary and operational, if equipped. A locomotive may continue in lead with a defective or unsanitary toilet facility when all of the following conditions are met:
 - The unsanitary or defective condition is discovered at a location where there are no other suitable locomotives available for use.
 - The locomotive, while non-compliant, did not pass through a location where it could have been cleaned, repaired, or switched with another compliant locomotive since its last daily inspection.
 - Upon reasonable request of a crewmember operating with a defective or unsanitary toilet facility, arrangements are made for the crew to access a sanitary toilet facility outside of the locomotive.

5.5.2 Walkway, Car Body, and Engine Compartment

From the walkway, car body, and engine compartment, inspect both sides of each locomotive to ensure:

- Walkways and walk-in compartments are clear of debris, tools, and accumulated oil or grease.
- Handrails, handholds, steps, ladders, safety chains, and guards are secured and ready for service:
 - Inspect for broken, bent, damaged, or loose equipment.
 - Make sure safety chains are connected high enough for safe passage.
 - No passageway is required through the nose of a car body locomotive.
 - Locomotives with open platforms not coupled to another locomotive must have a continuous barrier across the full width of the end of a locomotive.
 - Locomotive with an open platform coupled to another locomotive without an open platform must have a continuous barrier between locomotives.
- All electrical and rotating equipment guards are in place.
- The diesel engine has no apparent exhaust, oil, water, or fuel leaks.
- Locomotive hand brake is operational.

5.5.3 Ground Inspection

From the ground level, inspect both sides of each locomotive to ensure:

- Sand is deposited on the rail in front of the lead wheels on each locomotive in consist.
 - A locomotive being used in switching service at a location not equipped with a sand delivery system may continue in service for seven calendars from the date the sanders are found to be inoperative or the next periodic inspection, whichever occurs first.
- The fuel tank is not leaking.
- No defects such as cracked, broken, or missing parts are on the following:
 - Locomotive trucks.
 - Wheels.
 - Draft gears.
 - Gear cases.
- Brake cylinder piston travel must be as follows:
 - Minimum: Sufficient to provide brake shoe clearance when the brakes are released.
 - Maximum: 1 ½ inches less than the travel entered on FRA Form F 6180-49A (blue card).
- Foundation brake rigging is secured and all components other than wheels and sand hoses are at least 2 ½ inches above the top of the rail.
- Brake shoes are secured and approximately in line with the tread of the wheel and there are no obvious lips or overhangs.
- No part of the electrical jumper cable is lying on the coupler.
- Unused electrical cables must be stowed properly or the disconnected ends placed into a dummy receptacle or a multiple-unit cable holder.
- Cable and cable connections between locomotives may not have broken or badly chafed insulation, broken plugs, receptacles, or terminals, or broken or protruding strands of wire.
- The coupler does not have a crack or break in the pulling face of the knuckle.
- Snowplow, pilot, or end plate is properly, secured and is between 3 and 6 inches above the top of the rail.
- Main reservoirs are drained of oil and water at least once per day.
- Automatic drain valve handles are turned fully clockwise to the automatic position with the stem extending beyond the valve handle, if equipped.

5.6 Recording Daily Inspection

A written report of the inspection shall be made on the Locomotive Daily Inspection Report or other equivalent foreign carrier form. The following information must be included:

- Time.
- Location.
- Description of non-complying defects.
- Signature of Inspector.
- Non-complying repairs:
 - Description of repairs.
 - Signature of person making repairs.

5.7 Tagging Non-Complying or Defective Locomotives

When a non-complying condition or other defect is identified, prior to moving the locomotive(s), complete the following:

- Report condition or defect to Mechanical and other proper authority.
- Record the required information on a locomotive defect tag.
 - Identify defect type by checking the applicable box (non-complying or defect).
 - Locomotive initials and number.
 - Name of the inspecting railroad.
 - Inspection location and date.
 - Nature of defect(s).
 - Movement restrictions, if any.
 - Destination.
 - Signature of the employee making the inspection.
- Attach the locomotive defect tag to the isolation switch of the non-complying or defective locomotive.
- Attach a copy of the locomotive defect tag on the control stand of the controlling locomotive.
- The locomotive engineer in charge of the movement must receive written notification of the non-complying locomotive; a copy of a locomotive defect tag meets this requirement.
 - The locomotive engineer must inform all other crew members of the non-complying locomotive and of any restrictions.

A locomotive defect tag is not required provided the locomotive is being moved as a single locomotive or dead in a yard for the purpose or repair at a speed not exceeding 10 mph.

5.8 Movement of Non-Complying Locomotives

5.8.1 Locomotives Safe to Move

During the locomotive daily inspection if one or more non-complying defects are identified, a locomotive may be moved if it is determined by a qualified person the locomotive is safe to move. If the locomotive is safe to move, it may only be moved:

- As a single unit under power not attached to cars.
- In a locomotive consist not attached to cars.
- Isolated or shut down when attached to cars.
- After a locomotive defect tag is completed and attached to the isolation switch.

Exceptions:

- A controlling locomotive with a defective speed indicator identified during a daily inspection may be operated under power attached to cars not exceeding 20 mph.
- Locomotive(s) found with the following defects during the daily inspection may be operated under power attached to cars as a trailing locomotive:
 - Inoperative headlights or ditch lights.
 - Inoperative horn or bell.
 - Defective speed indicator.
 - Window cracks that obscure view.
 - Improperly secured cab seats.

- Inoperative safety devices.
- Defective or unsanitary toilet compartment.

5.8.2 Locomotives Not Safe to Move

During the locomotive daily inspection if one or more non-complying defects are identified and the locomotive(s) are not safe to move:

- Report the condition or defect to Mechanical and other proper authority.
- Complete a locomotive defect tag and attach as required.

5.8.3 Non-Complying Condition(s) Found Enroute

A locomotive that develops a non-complying condition enroute may continue in service provided the locomotive engineer or other qualified employee determines that the locomotive is safe to move. The locomotive may be operated until the next daily inspection or the nearest point where repairs can be made, whichever comes first. The locomotive engineer must:

- Report the condition or defect to Mechanical and other proper authority.
- Record the defect information on the Crew Information Form.
- Complete the locomotive defect tag and attach as required.
- Notify the relieving locomotive engineer of non-complying conditions.

5.8.4 Defects Other Than Non-Complying Conditions

If a defect other than a non-complying condition (bad order) is found during a daily inspection or while enroute, the locomotive engineer must:

- Report the condition or defect to Mechanical and other proper authority.
- Record the defect information on the Crew Information Form.
- Complete the locomotive defect tag and attach as required.

5.8.5 Major Internal Defects Found Enroute

If a locomotive develops a major internal defect enroute, comply with the following:

- Immediately shut down the engine.
- Report the condition to the train dispatcher and Mechanical.
- Do not restart the engine unless instructed by Mechanical.
- Complete the locomotive defect tag and attach as required.
- If instructed to set out the defective locomotive, when possible, leave in a location that can be accessed by Mechanical.

If safe to do so, a dead locomotive may continue as a controlling locomotive until the next daily inspection or the nearest point where repairs can be made, whichever comes first.

6.0 Locomotive Flat Wheel and Shell Spots

While inspecting locomotive wheels, measure and report as follows:

- Determine the length of the flat or shelled area.
- Notify the dispatcher and Mechanical when the flat spot or shelled area exceeds 1 inch.
- Use the table below to determine if equipment restrictions apply.

Locomotive Speed for Flat Wheels and Shell Spots		
Length of Single Spots	Length of Two Adjoining Spots Less Than 1-1/2" Apart	Maximum Speed
1" or less	1" or less	Maximum authorized speed.
1" to 1½"	1" to 1½"	40 mph
Over 1½ to 2"	Over 1½" to 2"	25 mph
Over 2"	Over 2"	10 mph; Set out locomotive at first available point but not before reporting to train dispatcher

7.0 Dynamic Brake Requirements

On train movements equipped with dynamic brakes, the controlling locomotive must:

- Have the ability to control the dynamic brakes on trailing locomotives in the consist, if equipped and operative.
- Be equipped with either an operative dynamic brake or accelerometer that displays current change in speed or predicted change in speed in miles per hour per minute.

8.0 Driving Axle Requirements

8.1 General Restrictions – All Trains

The following general driving axle restrictions will apply to all trains:

- No more than 28 driving axles are permitted in the lead consist of loaded crude oil trains.
- No more than 24 driving axles are permitted in the lead consist of trains handling DOT 111 tank cars.
- No more than 16 driving axles are permitted on trains operating over tracks where 6 axle locomotives are prohibited.

8.2 Driving Axle Limitations – Conventional Trains

Unless otherwise specified by this section or other instruction, a conventional train must not exceed 30 driving axles. The following exceptions apply:

TrAM Areas 1, 2, and 3

Trains are permitted to operate with up to 36 driving axles when the following conditions are met:

- The head 15 cars must weigh at least 50 tons.
- When multi-platform cars are part of the head 15 cars, no platforms can be empty and the average weight of all platforms on each car must be 50 tons or more.
- When energy management is active, manual control must be taken when tractive effort exceeds 100 Klbs as indicated by the lead locomotive.
- When operating at speeds below 12 mph, do not exceed the following tractive effort levels as indicated by the lead locomotive:
 - 135 Klbs when operating with 30 to 35 driving axles.
 - 110 Klbs for 36 driving axles.

TrAM Areas 4, 5, and 6

The following driving axle requirements will apply:

- Do not exceed 24 driving axles.

8.3 Driving Axle Limitations – Distributed Power Trains

The maximum number of driving axles permitted on distributed power trains are as follows:

TrAM Areas 1, 2, and 3

- Lead Consist: 30.
- In-train Remote(s): 12.
- Rear Remote: 12.

TrAM Areas 4, 5, and 6

- Lead Consist: 24.
- In-train Remote(s): 12.
- Rear Remote: 12.

Additional locomotives may be included in remote consists provided they are isolated.

The driving axle limits above may only be exceeded when an approved configuration is listed in the Distributed Power Placement chart. In the case of doubt or uncertainty, crew members must contact management for guidance.

8.4 Driving Axle Ratings

Driving axle ratings are as follows:

- 4 driving axles:
 - All 4 axle locomotives (except GP60).
- 6 driving axles:
 - GP60.
 - SD30.
 - SD40.
 - SD50.
- 8 driving axles:
 - D-8, D-9, D-9-40CW, D-944CW.
 - SD60.
 - SD70 (Non ACU models).
- 12 driving axles:
 - AC4400.
 - ES44AC.
 - SD70AC.
 - SD90.

When the driving axle rating of a locomotive is unknown, the following will apply:

- 4 axle locomotives – 6 driving axles.
- 6 axle DC locomotives – 8 driving axles.
- 6 axle AC locomotives – 12 driving axles.

9.0 Coupler Alignment Control

Unless otherwise specified:

- All leased locomotives will be equipped with coupler alignment control.
- All private industry locomotive(s) and the CP 6711 must be marshalled in-train.

Locomotives without coupler alignment must be marshalled as follows:

- As part of the lead consist:
 - Marshall immediately behind the lead locomotive.
 - If no locomotive is marshalled immediately behind the locomotive without coupler alignment control, the first two cars of the train must be no less than 65 feet and weigh at least 45 tons.
- As part of the train:
 - Comply with requirements for handling dead or disabled locomotives in a train.

10.0 Placement of Dead or Disabled Locomotives

10.1 Placing Dead/Disabled Locomotives Behind Lead Consist (Preferred Method)

Dead or disabled locomotives may be coupled directly behind the lead consist provided:

- Each locomotive is equipped with coupler alignment control.
- Locomotive(s) are conditioned using the dead engine cut out cock under the direction of the mechanical department (as required).
- Movements are governed by any restrictions and/or instructions placed by the Mechanical Department.

10.2 Handling Dead/Disabled Locomotives in Train (Alternative Method)

Dead or disabled locomotives, including locomotives without coupler alignment control, which cannot be placed within the lead consist, may be placed in-train provided:

- Passes TrAM check.
- Locomotive(s) are conditioned using the dead engine cut out cock under the direction of the Mechanical department (as required).
- No more than two dead/disabled locomotives are marshalled in-train.
- Movements are governed by any restrictions and/or instructions placed by the Mechanical department.

A single locomotive with a damaged drawbar must be positioned to the extreme rear of a train and may be moved provided:

- Passes a TrAM check.
- The locomotive is conditioned using the dead engine cut out cock under the direction of the Mechanical department (as required).
- Air brakes are operative.
- Movements are governed by any restrictions and/or instructions placed by the Mechanical department.

11.0 Cold Weather Operation – High Idle

When the ambient temperature is expected to be at or below 0° Fahrenheit, prior to being left unattended, locomotives not equipped with high idle protection must be throttled up manually.

To manually throttle up a locomotive:

- Generator Field Off.
- Control/Fuel Pump and Engine Run On.
- Reverser Centered.
- Throttle advanced to notch 4.

12.0 Draining a Locomotive

During cold weather operation, when a locomotive is shut down and cannot be restarted, the engine cooling water must be drained as outlined by the draining instructions posted in the cab. When in doubt, contact Mechanical for guidance.

When draining engine cooling water, ensure all people are clear of the cooling water drainpipe pressure relief outlet and the filler pipe connector.

13.0 Starting and Stopping Locomotives

Procedures to start and stop locomotives are posted within the locomotive cab or starting station. If required, contact Mechanical for assistance.

14.0 Isolating Locomotives

The following will apply when isolating locomotive(s):

- While in motion, the isolation switch of a lead locomotive must be in the run position at all times.
- Unless otherwise specified, light engine consists may only operate with one locomotive on-line.
- While a locomotive is under load, do not place the isolate switch to run or isolate.

15.0 Response to Alarms

Locomotive alarms must be investigated as soon as possible. If a continuous wheel slip, pinion slip, or locked axle indicator is displayed, stop the movement immediately and contact Mechanical for instructions.

16.0 Crankcase Overpressure

Do not restart a locomotive that has shut down due to crankcase overpressure. Contact Mechanical for guidance.

17.0 Locomotive Power Limit Switch

If equipped, the power limit switch (HP Boost) must be on at all times.

18.0 Snowblaster and Rail Cleaner Operation

On locomotives equipped, compressed air is used to clean snow and debris from the rail. The cleaning device is installed near each lead axle sand bracket on the front of the locomotive and may be activated automatically or manually.

19.0 Fire Prevention

19.1 Preventing Sparks

Locomotives that idle for extended periods of time may discharge sparks from the exhaust stack. To reduce the possibility of sparks on stationary locomotives that have been idling for extended periods, perform the following:

- Center the reverser.
- Advance the throttle to notch 5 for 10 minutes.

While operating, train crews must be alert to evidence of excessive spark emission from the locomotive(s) or train.

19.2 Locomotives Suspected of Starting Fires

Whenever possible, shut down locomotive(s) that are suspected of starting fires. This information must be recorded on the Crew Information Form and reported to Mechanical.

20.0 Electronic Display Screen Failure on Lead Locomotive

On conventional trains, if one display screen fails, the locomotive may continue in service. If both screens fail, the locomotive must be repaired or removed from the lead position. On distributed power trains, if one or both screens fail, the locomotive must be repaired or removed from the lead position.

All screen defects must be promptly reported to Mechanical.

21.0 Event Recorder and Camera Systems

Access to event recorder and camera systems is restricted. Only authorized personnel may extract event recorder and camera data. The camera system must be on and unobstructed at all times. Any action which interferes with the normal operation of the event recorder or camera system will be subject to discipline.

22.0 Locomotive Safety Devices

Locomotive safety devices must be cut in and operating at all times.

This requirement does not apply to the following:

- Non-controlling locomotives.
- Distributed power locomotive(s).
- Dead-in-tow locomotives.

Safety device malfunctions must be immediately reported to Mechanical and other proper authority. Do not cut out, tamper with, or defeat a safety device without proper authorization.

23.0 Alerters

23.1 Alerter Requirements

All controlling locomotives that operate at speeds greater than 25 mph must be equipped with an operative alerter. If the alerter malfunctions enroute, then the locomotive engineer must:

- Immediately notify the dispatcher and document the malfunction on the Crew Information Form.
- Contact and be governed by Mechanical's instructions for cut out of the alerter.

A locomotive with an inoperative alerter may continue to operate as a controlling locomotive until the next daily inspection or forward point where repairs can be made.

23.2 Test Procedure

To test the alerter on a locomotive, complete the following:

- Protect the locomotive(s) from unintended movement.
- Initiate a penalty brake application by placing the automatic and independent brake handles in the release position and allowing the alerter to time out.
- Ensure the audible alarm and warning lights are functioning properly.
- Ensure equalizing reservoir pressure reduces to 61-psi or lower.
- Recover the penalty brake application.

23.3 Alerter Nullification

When required to temporarily nullify the alerter for loading and unloading unit bulk trains, the following will apply:

Standard Procedure

These steps will nullify the alerter at speeds of 2 mph or less:

- Move throttle to Idle.
- Center reverser.
- On distributed power trains, place remote consist(s) in remote mode Idle.
- Isolate the controlling locomotive.
- Release the independent brake.

Alternate Procedure (to be Used When Standard Procedure is Not Effective)

These steps will nullify the alerter at speeds of 4 mph or less:

- Isolate all locomotive in the lead consist, except the controlling unit.
- On distributed power trains, place remote consist(s) in remote mode Idle.
- Select slow speed control on the operator screen.
- Set slow speed control to zero or lowest speed possible if zero is unavailable.
- Move the reverser to the direction of travel.
- Advance throttle to notch 1.
- Isolate lead locomotive.
- Release the independent brake.

24.0 Locomotive Speed Indicators

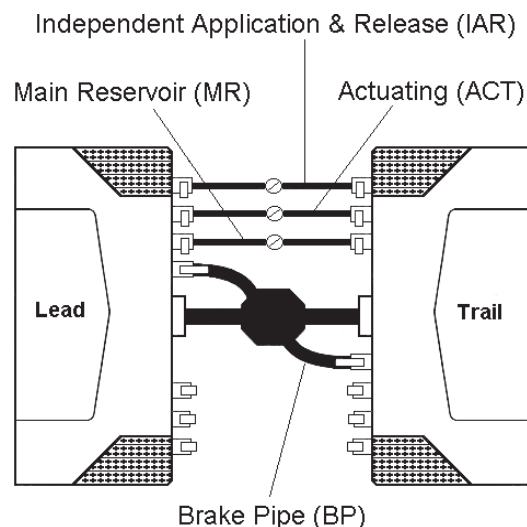
A locomotive used as a controlling unit at speeds above 20 mph must be equipped with an operative speed indicator. If a speed indicator on a controlling locomotive fails enroute, it must be handled as a non-complying condition found enroute. The locomotive may continue in the lead position at maximum authorized speed until the next calendar day inspection or next repair facility, whichever occurs first.

The speed indicator must be tested for accuracy as soon as possible after leaving a terminal using measured miles or other equivalent procedures. Speed indicators must be accurate to ± 3 mph at speeds of 10 to 30 mph and within ± 5 mph at speeds above 30 mph. When an inaccuracy of the speed indicator is observed, train speed must be adjusted accordingly, and the exception must be noted on a Crew Information Form. Report the occurrence to the train dispatcher.

25.0 Coupling Locomotives for Multiple Unit (MU) Operation

To connect locomotives for multiple unit operation:

- Connect the electrical jumper cable between locomotives.
- Couple the corresponding multiple unit hoses.
- Open brake pipe and multiple unit hose angle cocks between locomotives.
- Connect walkway safety chains and lower walkway platform, if equipped.



26.0 Separating Locomotives

When separating locomotives:

- Disconnect walkway safety chains and position the walkway platform, if equipped.
- Disconnect MU cables.
- Plug the MU cables into a dummy receptacle or store in other designated locations.
- Close cut out cocks and angle cock.
- Attach air hoses to the dummy couplings or place them in the pockets.

27.0 Locomotive Air Brake Test

27.1 Locomotive Air Brake Test Requirement

Perform a locomotive air brake test when:

- A locomotive consist is first assembled or coupled together.
- Adding locomotive(s) to a consist.
- Removing other than rear locomotive(s) from a consist.
- Changing operating ends.
- The locomotive consist is rearranged.

Locomotive air brake test is not required when taking power from a mechanical facility with a completed Locomotive Consist Information Form or notified the test has been previously performed.

27.2 Performing a Locomotive Brake Test

To perform a locomotive air brake test, complete the following:

- Protect the locomotive(s) from unintended movement.
- Ensure a qualified person is positioned on the ground to confirm that all brake cylinders extend and retract as intended on the locomotive(s) being tested.
- Place the automatic brake handle in the release position for at least two minutes to ensure the locomotive air brake system is fully charged.
- With the independent and automatic brake handles in release, fully apply the independent brake and verify all locomotive brakes apply.
- Release the independent brake and verify all locomotive brakes release.
- Make a 15-psi automatic brake reduction and verify all locomotive brakes apply.
- Actuate the independent brake for at least four seconds per locomotive and verify all locomotive brakes release.
- Make a further reduction with the automatic brake, no less than full service, and verify all locomotive brakes apply.
- Release the automatic brake and verify all locomotive brakes release.
- Test the alerter, except when adding trailing locomotive(s).

28.0 Moving Locomotive(s)

28.1 Prior To Movement

Prior to making an initial movement of a locomotive consist, make a visual inspection to verify:

- Locomotive air brakes are applied on each locomotive.
- Required air hoses are coupled between all locomotives in the consist.
- Cut out cocks and valves are positioned correctly.
- Operational status of locomotives.
- Main reservoir pressure is higher than 105-psi and air brake pressures are within standard.
- Hand brakes are released on all locomotives.

When locomotive status has been documented on a Locomotive Consist Information Form located on the lead consist, the locomotive engineer is relieved from performing a visual inspection of the first five bullet points listed above.

28.2 Changing Operating Ends

When changing operating ends or during the initial movement of a locomotive consist, perform the following:

- After movement begins and speed is below 10 mph, apply the independent brake.
 - Ensure brake cylinder pressure develops and speed reduces.
- Release the independent brake and ensure the locomotive(s) roll freely.
- Make a 10-psi reduction with the automatic brake.
 - Ensure brake cylinder pressure develops and speed reduces.
- Actuate the independent brake and ensure the locomotive(s) roll freely.

28.3 Operating Cab Requirements

Operate a light locomotive consist from the cab nearest the direction of travel when any of the following apply:

- The distance to be traveled exceeds two miles.
- A member of the crew does not control the movement using hand signals or radio.
- Visibility is impaired.

28.4 Moving Locomotive Consists in a Terminal

Multiple locomotive consists may be moved within a terminal area with only the brake pipe connected provided:

- Movement does not exceed 10 mph.
- The automatic brake is used to control speed.
- Brake pipe is connected and angle cocks are open between locomotives.
- All locomotives coupled to the controlling locomotive are conditioned properly:
 - Automatic brake valve(s) cut out and in handle off position.
 - Independent brake valve(s) in trail and in release position.
- Brake test is performed:
 - Make a 10-psi automatic brake reduction.
 - Ensure brakes apply on all locomotives.
 - Release the automatic brake.
 - Ensure brakes release on all locomotives.
- Release all hand brakes.

28.5 Hostling Locomotives

When hostling locomotives with inoperative brakes, a minimum of one locomotive with operative brakes must be used per six locomotives with inoperative brakes.

29.0 Modifying Operating Controls

29.1 Cutting Out Operating Controls

To cut out operating controls:

- Ensure locomotive(s) are secured against unintended movement.
- Move the independent brake handle to full application.
- Make a 20-psi reduction with the automatic brake.
- Center the reverser and remove the handle.
- Place the generator field switch in the off position.
- Disarm the EOT, when required.
- Position headlight switch, if necessary.
- Cut out the automatic brake.
- Move the automatic brake handle to handle off / continuous service.
- Cut out the independent brake.
- Move the independent brake handle to release and ensure brake cylinder pressure is maintained.
- If changing operating controls, immediately proceed to the locomotive to be cut in.

29.2 Cutting In Operating Controls

To cut in operating controls, condition the control stand as follows:

- Move the independent brake handle to full application.
- Cut in the independent brake.
- Move the automatic brake handle to release.
- Check air gauges and verify pressures are within standard.
- Cut in the automatic brake.
- Place the generator field switch in the On position.
- Ensure the engine run switch is in the On position.
- Ensure the control/fuel pump switch is in the On position.
- Conduct required air brake test.

30.0 Traction Motors

30.1 Traction Motor Auto Cut Out

When the locomotive screen indicates that a traction motor has automatically cut out, perform the following:

- Report the defect to Mechanical.
- Ensure that all traction motor speed sensors are cut-in/enabled.
- Document defect(s) and speed sensor status on the Crew Information Form.

30.2 AC Traction Motor Failure

- Do not cut out traction motors or speed sensors without authorization from Mechanical.
- When a potential AC traction motor failure is identified, stop the movement and inspect for the following:
 - Wheel skids or shelling.
 - Electrical burn smell.
 - Any signs of heat damage verified by application of a temperature indicating crayon on the journal bearings.
 - Gearcase leaks.
 - Unusual noises from traction motors.
- Following inspection, contact Mechanical. Report any defects noted above and be governed by instructions received. Document defects, speed sensor status, and Mechanical instructions the Crew Information Form.

30.3 DC Traction Motor Failure

- When a potential DC traction motor failure is identified, stop the movement, and inspect for the following:
 - Wheel skids or shelling.
 - Electrical burn smell.
 - Any signs of heat damage verified by application of a temperature indicating crayon on the journal bearings.
 - Gearcase leaks.
 - Unusual noises from traction motors.
- Following inspection, contact Mechanical. Report any defects noted above and be governed by instructions received. All defects and Mechanical instructions shall be documented on the Crew Information Form.
- If a DC traction motor is cut out:
 - Locked axle detection function is disabled must be continually monitored to ensure the wheels are rotating freely.
 - Ensure the locomotive is marshalled in the lead consist for observation.
 - The locomotive must be set off at the nearest set out point, not to exceed the next daily inspection or forward point where repairs can be made, whichever occurs first.

30.4 Temporary Wheelsets

When a defective traction motor assembly has had the pinion gear cut or has been replaced with a temporary wheelset, the traction motor and speed sensor are to be manually cut out for only that position. The locomotive must be marshalled in the lead consist and closely monitored to confirm that the wheels are rotating freely and are not making any unusual noises. Mechanical will advise of any speed or operating restrictions. The Crew Information Form must be updated accordingly.

31.0 Speed Sensors

31.1 Speed Sensor Failure

If Mechanical has confirmed that a traction motor speed sensor failure requires the speed sensor to be manually cut out, the locomotive must be marshalled within the lead consist. The identified wheel must be continually monitored to confirm that it is rotating freely and not making any unusual noises. The locomotive must be set off at the nearest set-off point. The Crew Information Form must be updated accordingly.

32.0 Distance Measuring Device

32.1 Distance Measuring Device Requirements

The distance measuring device, also referred to as a counter, allows train crews to determine the location of the rear car of a train in relation to a given reference point where the counter is activated. When programming the distance measuring device, enter the actual train length of the train. In non-signaled territory, add an additional three percent to the total train length. The train length entered into the distance measuring device must be updated when the train consist or locomotive(s) are changed.

Crews must exercise discretion and travel an extra distance if a stopping location, clearance point, signal, or other restriction requires. Distance measuring devices must be checked for accuracy against known distances, which may include predetermined measured mile locations or other locations specified by instruction.

Section 4 – Securing Equipment

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1.0 General Requirements

Crew members are responsible for securing unattended equipment with hand brakes. Equipment is considered unattended when left in a manner that the brake system of the equipment cannot be readily controlled by a qualified person. Employees must not engage in any unrelated tasks while attending equipment.

2.0 Hand Brakes

Fully apply hand brakes until the slack is out of the chain and the brake shoes are in contact with the wheel. In winter conditions, ensure that braking surfaces are free of ice and snow.

Unless testing for securement, hand brakes must be fully released before moving equipment.

3.0 Testing Hand Brake Effectiveness

To ensure a sufficient number of hand brakes are applied, release all air brakes and allow sufficient time for release. Cause or allow the slack to adjust to verify that the number of hand brakes applied are sufficient to prevent equipment from moving. This test must be completed before uncoupling or leaving equipment unattended.

When unable to test hand brake effectiveness, apply hand brakes as follows:

- On grades from 0 to 0.99%, apply hand brakes to at least 20 percent of cars.
- On grades from 1.0 to 1.29%, apply hand brakes to at least 30 percent of cars.
- On grades from 1.3 to 1.79%, apply hand brakes to at least 50 percent of cars.
- On grades 1.8% or higher, apply hand brakes to at least 65 percent of cars.
- Locomotive hand brakes must not be included in the total hand brake count. If in doubt or uncertainty exists as to the number of hand brakes to apply, contact management for instructions.

4.0 Securing Equipment Following an Emergency Brake Application or Train Braking Issue on Heavy or Mountain Grade

When a train experiences an emergency brake application or train braking issue on heavy or mountain grade, the following will apply:

- Immediately secure equipment with hand brakes prior to attempting to recharge the air brake system.
- At minimum, apply hand brakes in accordance with the following:
 - On grades from 1.0 to 1.29%, apply hand brakes to at least 30 percent of cars.
 - On grades from 1.3 to 1.79%, apply hand brakes to at least 50 percent of cars.
 - On grades 1.8% or higher, apply hand brakes to at least 65 percent of cars.
- Once the air brake system is fully recharged, prior to the release of hand brakes, make an automatic brake application sufficient to hold the train.

5.0 Leaving Equipment Unattended

5.1 General Requirements

When leaving railway equipment unattended, the following will apply:

- A single piece of equipment must be left with the hand brake applied.
- A single piece of equipment with a defective hand brake must be left coupled to equipment with an operative hand brake.
- Two or more pieces of equipment must be left with a sufficient number of hand brakes.
- When hand brakes are not adequate, block the wheels.
- When it is required to remove equipment from a previously secured track or train, ensure the equipment left standing is properly secured with hand brakes.
- Do not uncouple from or allow equipment to move under its own momentum into standing equipment unless it is known that the number of hand brakes applied on the standing equipment are sufficient to prevent movement of all equipment.

5.2 Leaving Locomotive(s) Unattended

To secure unattended locomotive(s), perform the following:

- Apply hand brakes to each locomotive.
- Test hand brake effectiveness.
- Condition one control stand as a lead locomotive.
 - Independent brake cut in and fully applied.
 - Automatic brake cut in and 20-psi reduction.
 - Generator Field Off.
 - Engine Run and Control / Fuel Pump On.
 - Engine Control Switch to Isolate.
 - Reverser handle removed.
- Turn off all unnecessary lights and close all windows.
- Lock all doors on each locomotive, unless being left in one of the following yards or terminals with 24 hour per day oversight:
 - Artesia, Beaumont, Bensenville, Heavener, Jackson, Kansas City, Kendleton, Laredo, Mason City, Meridian, Monroe, Mossville, Nahant, Pittsburg, Saratoga Springs, Shreveport, St. Louis, St. Paul, or Wylie.

5.3 Leaving a Portion of a Train or Cars Unattended

To secure a portion of a train or cars being left unattended, perform the following:

- Apply a sufficient number of hand brakes.
- Test hand brake effectiveness.
- Make an automatic brake reduction sufficient to prevent movement.
- The locomotive engineer must advise the crewmember when the angle cock can be closed.
- Cut away from the equipment and ensure an emergency brake application occurs.
- Leave the angle cock fully open.
 - The angle cock may be closed for the purpose of immediately running around equipment to open an angle cock or couple to the other end.

- Crew members are responsible to verbally confirm that equipment is being left in accordance with these instructions.

5.4 Leaving a Train Unattended

To secure an unattended train, perform the following:

- Apply a sufficient number of hand brakes to the car(s) in the train.
- Test hand brake effectiveness.
- Apply hand brakes to each locomotive in the lead consist.
- Leave the lead locomotive of the unattended train conditioned as follows:
 - Independent brake cut in and fully applied.
 - Automatic brake cut in and 20-psi reduction.
 - Generator Field Off.
 - Engine Run and Control / Fuel Pump On.
 - Engine Control Switch to Isolate.
 - Reverser handle removed.
- Turn off all unnecessary lights and close all windows.
- Lock all doors on each locomotive, including remotes.

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1.0 General Instructions

1.1 Locomotive Engineer Responsibilities

Locomotive engineers must be familiar with current information regarding locomotive systems and mechanical procedures and are responsible for the equipment in their care.

Locomotive engineers must employ proper train handling techniques to:

- Prevent damage to track structure, equipment, and lading.
- Utilize the most fuel-efficient means of operation consistent with good train handling.
- Ensure safe, efficient, and productive operation.

Proper train handling will depend on various factors. Locomotive engineers must consider the following while pre-trip planning and operating a train:

- Train marshalling.
- Train tonnage and length.
- Tons per operative brake.
- Speed.
- Grade.
- Temperature and weather conditions.
- Amount of slack in train.
- Throttle response characteristics of locomotive consist(s).
- Available throttle and dynamic braking effort.
- Dynamic brake type and limitations.
- Number and placement of cushioned drawbars in the train.
- Block signal spacing.

1.2 Coupling Portions of a Train

When coupling together two portions of a train, a brake pipe reduction of 35-psi must be made before opening the angle cock, unless train movement can be prevented with the locomotive brakes.

1.3 In-Train Forces

Controlling and limiting in-train forces are essential for safe train operation. Unless an emergency or other condition requires immediate speed reduction, change throttle and dynamic brake positions slowly and utilize split reductions with the automatic brake to allow train slack adequate time to adjust.

1.4 Train Separation and Stall Requirements

1.4.1 General Requirements

Train stalls and separations must be reported to the train dispatcher and territory Road Foreman. Where possible, broken knuckles must be captured or placed in a location where they can be retrieved by management. Locomotive engineers are required to complete and submit required train stall and separation report form(s).

1.4.2 Responding to a Train Stall

Upon stall, the locomotive engineer must:

- Maintain throttle in position 5 or higher.
- Make a 15-psi automatic brake reduction.
- Monitor gauges and surroundings to ensure the train does not roll backwards.
- Once confirmed that the train is stationary and the automatic brake reduction is sufficient to hold the train, the throttle can be slowly reduced to Idle.
- If the train begins to move backwards:
 - Increase the throttle.
 - Increase the automatic brake reduction.
- If movement continues:
 - Rapidly move throttle to Idle, reverser to reverse, move handle to DB 8, place the train in emergency.
 - Secure the train in accordance with securement rules.

1.5 Winter Conditions

When snow or ice buildup is present, operate locomotive(s) through yard and industry tracks, or crossings within these tracks, prior to moving equipment. Exercise extreme caution when making movements through crossings with minimal rail activity or high levels of vehicular traffic.

2.0 Use of Independent Brake

The following requirements apply while using the independent brake:

- The independent brake valve on the controlling locomotive must remain cut in at all times and the handle must not be blocked in the actuate position.
- When it is desired to prevent the locomotive brakes from applying during an automatic brake application, the independent brake valve handle must be actuated prior to the automatic brake application and held until exhaust ceases, but no less than four seconds per locomotive.
- Except when starting or stopping a train or operating with light locomotive(s), the independent brake must not be applied while in power or dynamic brake.
- The independent brake may be used in conjunction with dynamic braking, only at speeds of 10 mph and lower.
- Light independent brake, not to exceed 15-psi, may be used to control wheel slip at speeds below 10 mph.
- When conditions require the independent brakes to be applied, brake cylinder pressure must be controlled to prevent overheating or sliding of the locomotive wheels, excessive slack action, and high in-train forces.
- The independent brake must be fully applied when locomotive(s) are stationary.

3.0 Use of the Automatic Brake

3.1 General Requirements

The following general requirements apply to the use of air brakes:

- Use of the regulating valve to make brake pipe reductions is prohibited.
- If brake pipe pressure is reduced below 48-psi while making service brake applications, the train must be stopped and recharged.
- A train must not be started until the air brakes are fully released.
 - In the absences of brake pipe flow indication or rear car pressure reading, allow one minute for every 25 cars in the train.
- When possible, wait at least 30 seconds between split reductions.
- Use the automatic brake to control a light locomotive consist of seven or more locomotives. Do not bail the locomotive brakes unless a release is required.

3.2 Applying or Reapplying the Automatic Brake

When applying or reapplying the brakes, use the equalizing reservoir to measure reductions as follows:

Conventional Trains

- When the train air brake system is fully charged, make at least a 7-psi reduction.
- When the train air brake system is not fully charged, use one of the following methods to apply the automatic brake:
 - Determine the amount of false gradient in the train and reduce the equalizing reservoir pressure 7-psi plus the amount of false gradient (True Gradient – False Gradient + 7-psi).
 - Make an automatic brake reduction at least 7-psi below the rear car brake pipe pressure.
 - Make at least a 7-psi reduction below the point where brake pipe pressure begins to exhaust.
 - When a train equipped with an EOT does not display rear car pressure, make an automatic brake reduction of at least 5-psi more than the last reduction.

Distributed Power Trains

- When the train air brake system is fully charged, make at least a 7-psi reduction.
- When the train air brake system is not fully charged, use one of the following methods to apply the automatic brake:
 - When the train has only one remote consist which is located on the rear of the train, make at least a 7-psi reduction below the point where brake pipe pressure begins to exhaust.
 - When the rear car pressure is known, determine the amount of false gradient in the train and reduce the equalizing reservoir 7-psi plus the amount of false gradient.
 - When the train has multiple remotes with at least one remote consist located on the rear of the train, or a train equipped with an EOT that does not display rear car pressure, make an automatic brake reduction of at least 5-psi more than the last reduction.

- If the automatic brake valve is moved to the full service position and an additional automatic brake reduction is required to control train speed, the train must be immediately stopped and recharged. Hand brakes may be required to secure the train while the brake system is recharged.

3.3 Releasing the Automatic Brake

At slow speeds, evaluate the following conditions before attempting a running release of the automatic brakes:

- Train speed.
- Train makeup.
- Physical characteristics of the territory.
- Location of remote consist(s) and recharge characteristics.
- Train length.
- Brake pipe leakage.

Following a release of the automatic brake on trains equipped with an EOT, the throttle must not be increased, or dynamic brake reduced until the EOT pressure increases at least 3-psi and sufficient time has been allowed for the brakes to release.

On trains configured with a remote locomotive positioned at the rear of the train, the throttle must not be increased or dynamic brake reduced until the locomotive air flow indicates the brakes are released.

A running release may not be appropriate in all situations. Trains may be required to stop to comply with these instructions.

4.0 Train Braking Issues

4.1 Reporting Train Braking Issues

Immediately report the following to the train dispatcher:

- Any occurrence of an undesired release of the brakes.
- An automatic brake reduction of 20-psi or greater while moving.
- Abnormal flow or brake pipe gradient.
- A train that is not braking as anticipated.

Report must include the train's specific location, nature of the issue, current weather conditions, and any other relevant information.

4.2 Responding to Train Braking Issues

If an undesired release occurs while the automatic brake is applied, increase the automatic brake pipe reduction by at least 5-psi. The train may be placed in emergency, if necessary.

If unintended movement occurs while the train is stopped with air brakes set, increase the automatic brake pipe reduction by at least 5-psi. The train may be placed in emergency, if necessary. Once stopped, the train must be immediately secured with hand brakes.

Application of handbrakes will be governed by Securing Equipment Following an Emergency Brake Application or Train Braking Issue on Heavy or Mountain Grade in the Securement Chapter.

5.0 Delayed Departure in Non-Signal Territory

Anytime a train is stopped in non-signal territory, comply with the following:

Conventional Trains

- Make at least a 10-psi reduction with the automatic brake.
- When ready to depart, release the automatic brake.
- If brake pipe pressure is being restored, the train may proceed.
- If brake pipe pressure is not being restored, the train may operate up to one train length at a speed not to exceed 10 mph.
 - If the brake pipe pressure fails to restore during this time, the train must be stopped and inspected.
 - The train may depart when continuity is restored.

Distributed Power Trains

Complete a Train Check complying with related procedures.

Exceptions

The procedures listed above do not apply to conventional or distributed power trains when:

- A brief stop is made to line switches or comply with signal indications.
- On grades where train handling techniques require air brake to remain applied when initiating movement.

6.0 Minimizing Sticking Brakes

Sticking brakes occur when car air brakes remain applied after the automatic brake is released. When a sticking brake is identified:

- Stop the train as soon as possible, consistent with good train handling.
- Make a 20-psi reduction and allow sufficient time for air pressure to stabilize.
- If necessary, cut out the control valve or set out the car.

Utilize the following methods to minimize sticking brakes:

- Do not overcharge the air brake system.
- When the automatic brake is used to stop a train, make a reduction no less than 10-psi.
- When coupling cars to the rear end of a train, before opening the angle cock make a minimum reduction of at least 10-psi.
- When a running release of the automatic brake is to be made, if operating conditions permit, increase the automatic brake pipe reduction to at least 10-psi and allow the air brakes to stabilize before releasing.

7.0 Use of Throttle

7.1 Throttle Handling

When it is desired to increase speed, the throttle must be advanced one notch at a time. Tractive effort or amperage must be constant or dropping before moving the throttle to the next higher position. When it is desired to decrease speed, the throttle must be reduced gradually, except in emergency situations. The throttle must always be handled in a manner that results in smooth operation.

In addition:

- Do not change the generator field switch position while in throttle.
- Unless authorized to do so, do not utilize the throttle to hold a train stationary on a grade.
- After releasing the dynamic brake in preparation for applying power, the throttle must be advanced with care to ensure gradual adjustment of train slack.

7.2 Short Time Ratings

Short time rating limits apply to DC type locomotives when operating at high amperage levels. Do not exceed the short time ratings listed on the load meter or rating plate. Do not operate the locomotive continuously for more than the maximum time listed without stopping to cool the traction motors. Short time ratings are not cumulative.

When short time rating limitations exist and amperage levels are not displayed on the lead locomotive, the following will apply:

Speed

- 11 MPH – Continuous.
- 10.5 MPH – 60 Minutes.
- 10 MPH – 30 Minutes.
- 9 MPH – 15 minutes.
- 8 MPH – 10 minutes.
- 5 MPH – 5 minutes.

8.0 Use of Dynamic Brake

8.1 Dynamic Brake Factor

A dynamic brake factor is a numerical value used to convey the dynamic braking capability of a locomotive. The following dynamic brake factors will apply based on locomotive type:

Dynamic Brake Factors

- Dynamic Brake Factor 4:
 - All 4 axle locomotives (except GP60).
- Dynamic Brake Factor 6:
 - GP60.
 - SD30.
 - SD40.
 - SD50.

- Dynamic Brake Factor 8:
 - D-8, D-9, D-9-40CW, D-944CW.
 - SD60.
 - SD70 (Non ACU models).
- Dynamic Brake Factor 10:
 - AC4400.
 - ES44AC.
 - SD70AC.
 - SD80MAC.
 - SD90.

When the Dynamic Brake factor of a foreign locomotive is unknown, the following will apply:

- 4 axle locomotives – DB Factor 6.
- 6 axle DC locomotives – DB Factor 8.
- 6 axle AC locomotives – DB factor 10.

8.2 Dynamic Brake Limitations

Dynamic brake limits refer to the effort value, Klbs or amps, displayed on the lead locomotive gauge or screen. Do not exceed a dynamic brake factor of 20 on any locomotive consist. If the dynamic brake must be cut out on a locomotive without a cut out switch, the locomotive must be isolated.

The following will apply when the dynamic brake factor of the lead locomotive consist is 10 or greater:

- When operating on other than main track, or any track with a maximum speed of 15 mph or less, the DB effort must not exceed 60 Klbs (AC) or 500 amps (DC).
- When operating through any turnout, the DB effort must not exceed 60 Klbs or 500 amps (DC):
 - Conventional trains: Prior to the turnout, and until at least half the movement has passed the turnout.
 - Distributed Power trains: prior to the turnout, and until the entire movement has passed the turnout.
- When governed by a temporary speed restriction, the DB effort must not exceed 40 Klbs (AC) or 350 amps (DC):
 - Conventional trains: one-half mile prior to the restriction, and until the locomotive(s) are clear of the restriction.
 - Distributed Power trains: one-half mile prior to the restriction, and until the entire movement is clear of the restriction.
 - Exception: when any portion of such restriction is on a Heavy or Mountain grade, the DB effort must not exceed 60 Klbs (AC) or 500 amps (DC).
- When governed by a Heat Slow restriction, the DB effort must not exceed 60 Klbs (AC) or 500 amps (DC) until the entire movement is clear of the restriction.

8.3 Dynamic Brake Operation

The following requirements will apply to use of dynamic brakes:

- Pause in Idle for 10 seconds when transitioning from throttle to dynamic brake.
- If the wheel slip light illuminates while using the dynamic brake:
 - Reduce the dynamic brake handle position until the light goes out.
 - If the light continues to illuminate, stop the train and inspect the locomotives to ensure the wheels are rotating freely.
- When transitioning into dynamic braking, move the handle slowly to gradually bunch slack and minimize in-train forces.

9.0 Emergency Brake and Penalty Brake Applications

9.1 Location of Emergency Brake Valves

Employees must be familiar with the location and operation of emergency brake valves on locomotives and cars, if equipped.

9.2 Emergency Brake Application by Crew Member

A crew member must initiate an emergency brake application, when:

- Threat to life, personal injury, or property damage is imminent.
- The locomotive engineer becomes incapacitated or does not respond to warnings or signals to reduce train speed or stop the train.
- There is doubt or uncertainty that service brake applications can control train speed.

9.3 Exceeding Maximum Authorized Speed

When a train is descending a section of track defined as heavy or mountain grade and the train exceeds the maximum authorized speed by 5 mph or more, the train shall be immediately brought to a stop using an emergency brake application, if necessary.

9.4 Emergency Application and Undesired Emergency Procedures

To make an emergency brake application or respond to an undesired emergency application:

- Move the automatic brake valve handle to the emergency position.
- Activate the emergency toggle switch on the HOT if train is equipped with an EOT.
- While actuating, move the independent brake handle to a position that will develop the desired amount of brake cylinder pressure without sliding the wheels or developing excessive in-train forces.
 - Do not actuate when operating light locomotive consists.
- When the desired locomotive brake cylinder pressure is achieved, stop actuating the independent brake.
- To the extent possible, maintain the current slack condition to avoid the development of excessive in-train forces.
- If the emergency application occurs while in throttle, monitor gauges to ensure tractive effort reduces to zero. If tractive effort remains, move the throttle handle to idle.

9.5 Penalty Brake Application

When a penalty brake application occurs:

- If necessary, control the amount of independent brake cylinder pressure by moving the independent brake handle to the application zone while actuating.
- Move the automatic brake handle to the suppression position.
- Return throttle to Idle or dynamic brake to the off position.
- If operating conditions permit, after stopping and when the PCS closes, move the automatic brake handle to the release position.

9.6 Emergency Reversal of Locomotive Traction Motors

In emergency situations only, when all other means of stopping a train are exhausted, locomotive traction motors can be reversed. This procedure may be hazardous to personnel and equipment and should only be used as a last resort. If necessary to reverse the traction motors, comply with the following:

- Alert all employees on the locomotive.
- Place the throttle in Idle.
- Place the reverser handle in the position opposite the direction of travel.
- Advance the throttle to notch 1.

10.0 Shoving Movements

10.1 General Requirements

Use extreme care and good judgement while making shove movements. When beginning a shoving movement, use the lowest amount of tractive effort necessary to cause the equipment to move. To avoid jackknifing, wheel climb, or rail turnover, these factors must be considered before making a shoving movement:

- Long car/short car combinations and location in train.
- The number of empty cars near the head end of the train or remote consist(s).
- The amount of tractive effort produced.
- The number of locomotives that do not have alignment control couplers.
- Track curvature and grade.

10.2 Limitations While Shoving

The following limitations apply while making shoving movements and will be measured using the tractive effort or amperage value displayed on the controlling locomotive:

Maximum Allowable Tractive Effort		
AC Locomotives		
Locomotive Count	Operating Through Curves & Turnouts	Operating on Straight Track
1	150 Klbs	Maximum available
2	75 Klbs	110 Klbs
3	Prohibited	Prohibited
DC Locomotives		
1	900 amps	900 amps
2	450 amps	600 amps
3	300 amps	450 amps

If the above ratings are insufficient, amperage or effort may be increased, but only to the extent to cause the equipment to move. No more than two AC or three DC locomotives may be operating while making reverse movements. Additional locomotives must be isolated.

Each consist of a distributed power train are allowed the effort limits listed above provided each consist is separated by at least 35 cars or platforms.

11.0 Switching Movements

While switching cars, comply with the following:

- Gradually stretch or bunch slack when starting or stopping switching movements.
- Control buff and draft forces.

12.0 Operating Through Water

12.1 Response to Water Above the Top of Rail

Locomotives must not be operated, in throttle or idle, through water greater than 6 inches above the top of rail. This does not apply to rail cars provided movements are made in a push/pull fashion and locomotives are not moved through the water.

The following guidelines will govern response to water above the top of rail:

- A movement unexpectedly encountering water at any height above the top of rail must:
 - Make every effort to stop.
 - Immediately notify the train dispatcher.
 - Not proceed until the train dispatcher has confirmed with Engineering services that the water is 6 inches or less above the top of rail and the track structure is safe for movements.
- Should a movement unexpectedly operate through water at any height above the top of rail:
 - The movement must be stopped with the locomotives clear of the water.
 - The crew must immediately advise the train dispatcher and Mechanical department.

- The locomotive engineer must operate the traction motor blowers for 10 minutes per traction motor blower procedures.
- The movement must not proceed until the train dispatcher has confirmed with Engineering services that the water is 6 inches or less above the top of rail.

12.2 Traction Motor Blower Procedures

The following procedures will govern operation of traction motor blowers in response to water above the top of the rail.

EMD Locomotives with DC Traction Motors

- Ensure the locomotive is stopped with the reverser centered.
- Place the generator field switch to Off.
- Advance the throttle to notch 8 to increase the volume of air traveling through the traction motors.
- Confirm air is blowing from the traction motor vents.

SD70ACU, SD70ACE, SD70MACe Locomotives

- Ensure the locomotive is stopped with the reverser centered.
- On the main operator FIRE screen, Select Screen Controls.
- Navigate to the Locomotive Data page.
- Select Self Tests.
- Select the Traction Motor Blower test.
- Ensure all entry conditions for the test are met.
- Select Start and begin the test.
- Check the traction motor blower for abnormal sounds or vibrations.
- Notify Mechanical when abnormal conditions exist.
- Confirm air is blowing from the traction motor vents.
- After a minimum of 10 minutes, exit the test.

AC4400CW and C44-9W Locomotives

- Ensure the locomotive is stopped with the throttle in Idle.
- Open the cabinet doors behind the center seat.
- Toggle the Diagnostic Access Switch to gain access to the DID Panel.
- On the DID Panel, select Set, 2, Do_It.
- Select Test.
- Confirm the following by selecting Did_It.
- Air pressure.
- LCCB breaker is up.
- Disconnect all MU cables.
- Generator field breaker is up.
- COB breaker is up.
- Engine control switch is in Run.
- PCS is reset.
- Select Manual, New Test, Equipment Blower (Test 261 or 262), Do_It.
- Follow any set up instructions displayed on the screen.

- When all set up conditions are met, select Full Speed, then Do_It.
- Check the traction motor blower for abnormal sounds or vibrations.
- Notify Mechanical when abnormal conditions exist.
- Confirm air is blowing from the traction motor vents.
- Perform Half Speed and Quarter Speed Tests.
- Check the traction motor blow for abnormal sounds or vibrations.
- Notify Mechanical when abnormal conditions exist.
- Confirm air is blowing from the traction motor vents.
- After a minimum of 10 minutes, select Off and exit the test.

AC44CWM, ES44, and ET44 Locomotives

- Ensure the locomotive is stopped with the reverser centered.
- Contact Mechanical to obtain the current password to access level 3.
- From the operator's smart display screen, select Screen Controls, then Password.
- Enter the level 3 password.
- Navigate to the Loco Tests page.
- Select Self Test.
- Navigate to the Auxiliary Alternator page in the left column (Self Test group).
- Select Traction Blower Control (TBC) test in the Right Column or enter test number 406.
- Select Begin Test and wait for the auxiliary alternator to start up.
- Once the auxiliary alternator is running, select Full Speed.
- Check the traction motor blower for abnormal sounds or vibrations.
- Notify Mechanical when abnormal conditions exist.
- Perform Half Speed test.
- Check the traction motor blow for abnormal sounds or vibrations.
- Notify Mechanical when abnormal conditions exist.
- Confirm air is blowing from the traction motor vents.
- While the tests are running, confirm the amperage on the TBC (3 phases) rises evenly when the traction motor blower starts up.
- After a minimum of 10 minutes, exit the test.

13.0 Passenger Trains

13.1 Passenger Train Operations

When operating passenger trains, locomotive engineers are governed by the verbal and written instructions provided by the manager of the passenger train. These instructions will also apply to business trains.

14.0 Heavy and Mountain Grade Operation

14.1 General Requirements

The following handling requirements apply when cresting and descending a heavy or mountain grade. Heavy and mountain grade segments are identified in the timetable.

14.2 Handling guidelines

14.2.1 General Handling

- Train crews shall not operate down heavy or mountain grades where, in their judgement, a combination of ambient temperatures and heavy snow or ice conditions are sufficient to make the operation unsafe.
- Unless it is known that sufficient braking is available, when cresting hills towards heavy or mountain grade, maintain a speed at least 5 mph below maximum authorized speed until braking is determined to be ample.
- Early in the braking process, achieve a balance between the level of dynamic brake and the level of air brake needed to control train speed on descending grade.
- Where possible, limit automatic brake reductions to allow train speed to be controlled with the dynamic brake.
- If the brake pipe pressure at the rear of the train does not reduce in response to a reduction made with the automatic brake, stop the train immediately.
- When conditions warrant, use all available braking power including the emergency brake, if necessary.
- When stopped by an emergency application from any source:
 - The train must be immediately secured in accordance with securement rules.
 - The locomotive engineer must contact the proper authority and receive instruction from a Road Foreman if there is doubt or uncertainty that the train can be moved safely.
- Before moving from or recovering an emergency brake application on heavy or mountain grade, the crew must:
 - Consider train location, grade, proximity of lesser grade, weather, rail, or any other condition that may affect braking.
 - Contact the proper authority for instruction, should doubt or uncertainty arise.

14.3 Uncontrolled Movements

Any movement that reaches a speed of 5 mph above the maximum authorized tracks speed is considered an uncontrolled movement. Uncontrolled movements must be stopped immediately, using an emergency brake application, if necessary. After stopping an uncontrolled movement, a train may not proceed until it is determined that sufficient braking is available to control the movement.

14.4 Distributed Power Trains in Emergency with Communication Loss

When a distributed power train is standing in emergency on heavy or mountain grade and communication has been lost with one or more remote(s), perform the following:

- Secure the train with hand brakes in accordance with securement rules.
- Ensure the automatic brake handle of the lead locomotive is in emergency.
- End distributed power per instructions listed in Modifying or Ending Distributed Power Set Up.

15.0 Helper Operations

15.1 Use of Helper Locomotive(s)

Helper locomotive(s) may be used to assist a train provided there is not a caboose, riding platform, or crew transportation car at the rear of the train being assisted.

15.2 Operating Responsibilities

Comply with the following during helper operation:

- Exercise caution and good judgement at all times.
- When adding helpers to other locomotives on a train, control of all locomotives coupled together must be transferred to the lead locomotive engineer by plugging in the MU cable, whenever practicable.
- When more than one locomotive is attached to a train, unless specified in Helper Operation instructions, the locomotive engineer in the lead locomotive must control the train's air brakes.
- Prior to commencing helper operations, a job briefing must be conducted with all crew members to ensure understanding of:
 - Loads, empties, tonnage, and length.
 - Number of operative locomotives.
 - Dynamic brake factor.
 - Traction motor status.
 - Speed.
 - Equipment and movement restrictions.
 - Braking practices.
 - Where helper locomotive(s) will be removed.
- The locomotive engineer located on the leading end of a movement will be in charge of the following:
 - Movement Communication:
 - Limits of authority.
 - Signal indications.
 - Authorized speed.
 - Stopping locations.
 - All other information pertinent to the movement.
 - Providing warning with locomotive whistle and bell. This includes crossings at grade and other situations covered by rule.
- Radio communication between the assisted train and helper locomotive engineer(s) must be maintained at all times.
- The throttle of the helper locomotive must be advanced slowly, one notch at a time, maintaining the lowest possible effort to begin or maintain movement.
- Do not cut off helper locomotive(s) while the train is moving.

15.3 Placement of Helper Locomotives

15.3.1 General Requirements

The following will apply to the use of helper locomotive(s):

- The total number of driving axles at any location in the train may not exceed the driving axle limits listed in section 3.
 - Exception: No more than 24 driving axles may be used at the rear of any assisted train.
- A remote consist at the rear of the train must be included in the driving axle and locomotive count in helper operations.
- A TrAM check must be performed to ensure cars immediately ahead of the helper locomotive(s) pass Remote Zone rules when:
 - Assisting a train with distributed power consist(s) located in-train and no remote consist is located at the rear of the train.
 - Helper locomotives are placed in-train.
- Helper locomotives must adhere to established limitations for effort and amperage while shoving.
- While operating with independent control of throttle and brakes, the helper locomotive engineer must:
 - Follow all train handling commands from the locomotive engineer of the assisted train.
 - Immediately reduce the throttle to Idle and apply 50 percent of available independent brake if there is any indication that an emergency application has occurred while in motion.
 - Reduce the throttle to Idle, consistent with good train handling, if communication has been interrupted with the locomotive engineer of the assisted train and operational requirements are uncertain.

15.3.2 Helper Locomotive(s) Attached to the Head or Rear of a Train

The following requirements will apply to helper locomotives attached to the head or rear of a train.

Brake Pipe Disconnected

For movements that will travel less than three miles, helper locomotive(s) may assist a train with the brake pipe disconnected provided a job briefing is conducted with the locomotive engineer of the assisted train, and it is determined that the movement can be made safely with the brake pipe disconnected. If the movement cannot be safely made, follow procedures for brake pipe connected.

The helper locomotive engineer must:

- Obtain permission from the locomotive engineer of the assisted train to couple on.
- If coupling to the head end locomotive or distributed power locomotive:
 - Where possible, connect the electrical jumper cable between the helper locomotive(s) and the rear distributed power locomotive of the assisted train to allow throttle to be controlled by the locomotive engineer of the assisted train.
 - Place the automatic brake handle in release.
 - Remove the reverser handle.
 - Prior to commencing movement, fully release the independent brake.
- If coupling to a railcar:
 - Place the automatic brake handle in release.
 - Prior to commencing movement, fully release the independent brake.

Brake Pipe Connected

For movements that will travel a distance greater than three miles, control of the train's air brakes has changed, or where brake pipe must be connected, the following procedures will apply:

- The helper locomotive engineer will obtain permission from the locomotive engineer of the assisted train to couple on.
- Prior to opening angle cocks, the helper locomotive engineer must:
 - Make a 20-psi reduction with the automatic brake.
 - Cut out the automatic brake valve and place the handle in the handle off or continuous service position.
 - Couple the brake pipe hoses.
 - Open the angle cock on the helper locomotive.
 - Open the angle cock on the car or locomotive of the assisted train.
 - Place the independent brake valve handle in the release position and actuate to fully release the brakes of the helper locomotive(s).
- Once helper locomotive(s) are added to the train, the locomotive engineer of the assisted train must:
 - Perform a Class III air brake test.
 - Verify with the helper locomotive engineer that brake pipe pressure has reduced and increased in accordance with Class III requirements.

15.3.3 Helper Locomotive(s) Detached from the Head or Rear of a Train

After removing helper locomotive(s) from head or rear of an assisted train, perform the following:

- If the brake pipe was connected, the locomotive engineer of the assisted train must perform a class III brake test.
- If the brake pipe was not connected, no additional brake test is required.

15.3.4 Helper Locomotive(s) Marshalled into a Train

Perform the following when helper locomotive(s) are marshalled into a train:

- Prior to opening angle cocks, the helper locomotive engineer must:
 - Make a 20-psi reduction with the automatic brake.
 - Cut out the automatic brake valve and place the handle in the handle off or continuous service position.
 - Couple the brake pipe hoses.
 - Open the angle cocks on each side of the helper locomotive(s) and the assisted train.
 - Place the independent brake valve handle in the release position and actuate to fully release the brakes of the helper locomotive(s).
- Once helper locomotive(s) are added to the train, the locomotive engineer of the assisted train must:
 - Perform a Class III brake test.
 - Verify with the helper locomotive engineer that brake pipe pressure has reduced and increased in accordance with Class III requirements.

15.3.5 Removing Helper Locomotive(s) Marshalled into a Train

After removing helper locomotive(s) from an assisted train, the locomotive engineer of the assisted train must perform a Class III air brake test.

Section 6 – Fuel Conservation

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1.0 General Requirements

Employees must use the most efficient means available to maximize fuel efficiency. The following will apply:

- Locomotive engineers must use the most fuel-efficient methods when operating locomotives, consistent with good train handling.
- Dynamic brake is the preferred method of braking.
- Whenever possible, avoid increasing the throttle while the automatic brake is applied.
- Unless authorized by timetable or special instruction, power braking above notch 4 is prohibited.
- When it is known that a train will be delayed, train speed must be reduced to minimize delay time.
- Avoid unnecessary increases in speed over short distances.

2.0 Compliance with Train Document Messages and Other Instructions

The locomotive engineer and conductor are jointly responsible to ensure that the number of operating locomotives on a train are compliant with train document messages or other related instructions. Locomotive engineers must isolate excess locomotive(s) and document locomotive status on the Crew Information Form.

3.0 Energy Management Systems

When available, use of energy management software is mandatory. Locomotive engineers who have not received training are prohibited from operating energy management software.

Issues with energy management systems must be reported to the applicable support desk and reported on the required form.

4.0 Automatic Engine Start Stop (AESS)

4.1 General

Locomotives equipped with automatic engine start/stop systems are identified by labels and instructions affixed inside the locomotive cab and at the engine start/stop station. A green Enabled light will be visible on the engine control panel, or the control stand on certain automatic start/stop systems to convey system status. Warning horns or bells sound before an automatic shut down or restart occurs. Auto start/stop equipped locomotives will automatically shut down when conditions permit. The reverser must be centered for AESS to work on trailing locomotives.

4.2 Restarting Locomotives Prior to Starting a Movement

Locomotives equipped with automatic engine start and stop systems may have shut down if the locomotive(s) have been inactive for an extended duration of time. If it is suspected that any operative locomotive(s) has shut down, or if engine run status is unknown, move the reverser handle in the direction of travel to activate the engine start sequence. Wait at least two minutes before starting the train.

5.0 Locomotive Anti-idling Requirements

Locomotives not equipped with AESS system must be manually shut down if it is known that the locomotive(s) will be idling for more than 15 minutes, and the ambient temperature is expected to remain at or above 40 degrees Fahrenheit. In the case of uncertainty, contact Mechanical. The manual shut down requirement does not apply if the engine must be kept running to:

- Maintain air pressure for the braking system.
- Maintain battery charge on the locomotive for the starting systems.
- Heat or cool the cab or passenger equipment for the reasons of health and safety.
- Perform diagnostic testing and necessary maintenance.
- Respond to an emergency situation.

6.0 Pacing

When it is known that a train will be delayed, the train speed must be reduced to limit delay time and amount of fuel consumed. This includes following a slower train, anticipated train meets, and maintenance work windows.

The train dispatcher may issue instructions for train speed to be reduced to less than maximum authorized timetable speed. To take advantage of areas with descending grade, this restriction only applies when the train is operating in throttle.

When train pacing for an unusually long period of time, take additional locomotives off-line to compensate for reduced horsepower needs at lower speeds.

Delay Time Minutes	4	10	15	20	25	30	35	40	45	50	55	60
Distance from Delay Point (Miles)	SPEED REQUIRED IN MPH											
4 Miles	60	24	16	12	10	8	7	6	-		-	-
5 Miles	75	30	20	15	12	10	9	8	7	6	-	-
6 Miles	90	36	24	18	14	12	10	9	8	7	6	6
7 Miles	-	42	28	21	17	14	12	10	9	9	8	7
8 Miles	-	48	32	24	19	16	14	12	11	10	9	8
9 Miles	-	54	36	27	22	18	15	13	12	11	10	9
10 Miles	-	60	40	30	24	20	17	15	13	12	11	10
15 Miles	-	-	60	45	36	30	26	22	20	18	16	15
20 Miles	-	-	-	60	48	40	34	30	27	24	22	20
25 Miles	-	-	-	-	60	50	43	38	33	30	27	25
30 Miles	-	-	-	-	72	60	51	45	40	36	33	30
35 Miles	-	-	-	-	-	70	60	52	47	42	38	35
40 Miles	-	-	-	-	-	-	69	60	53	48	44	40
45 Miles	-	-	-	-	-	-	-	68	60	54	49	45
50 Miles									67	60	54	50

7.0 Trip Optimizer

7.1 General Requirements

The following requirements will apply to use of the Trip Optimizer system:

- Crews are responsible to ensure that the information entered into the Trip Optimizer system corresponds to the documentation for the train.
- Unless authorized to operate with distributed power locomotive(s) coupled to the lead consist, the locomotive engineer must conventionalize all locomotives located on the head end of the train. If authorized to proceed, the locomotive engineer must disengage Trip Optimizer or operate in manual mode and complete required reporting.

7.2 Trip Optimizer – Non-PTC Integrated (Stand-alone)

Non-PTC integrated Trip Optimizer plans a trip based solely on data received during the trip initialization. Trip Optimizer will not receive updates or have knowledge of critical operating information such as signals, track authority or restrictions that are voided or added during the trip. Non-PTC integrated Trip Optimizer is designated for main tracks only. Follow these procedures when:

- Initializing a new trip.
- Engaging Auto Control.
- Disengaging Auto Control.
- Using Auto Control Through Main Track Divergence.

7.2.1 Initialize New Trip

To initialize a new trip, perform the following:

- Press Trip Optimizer key.
- Press New Trip key.
- Press Yes key.
- Review trip information displayed and if correct, press Accept Trip key.
- Confirm by pressing Yes key.
- If trip information is incorrect, contact the applicable support desk.

7.2.2 Train Setup Verification

- Compare locomotive and train information displayed with train documentation received. If correct, press the Accept key.
- Confirm by pressing the Yes key.

7.2.3 Locomotive or Train Edits

If Locomotive or Train edits are required, changes can be accomplished as follows:

- Change Locomotives – To change the position of any locomotive, highlight that locomotive and use the number keys to enter the correct position.
- Remove Locomotive – To remove a locomotive, highlight the locomotive and press the Remove Locomotive key.

- Add Locomotive – Enter the initials and number of the added locomotive.
 - When locomotives are added, the default power mode is defaulted to isolate. The locomotive engineer must assign the correct operating status (Run, Isolate, DB Cutout, or DB Only).
 - The default position of the added locomotive(s) is 0.
 - A new position must be entered using the soft keys on the display screen to save changes to the locomotive consist.

On DP trains, it is critical that the position of the remote consist is accurately reflected on the Train Setup screen. Location is determined by line number, not loads/empties. Verify positioning and edit if necessary.

7.2.4 Change Power Mode

The Change Power Mode screen indicates the running status of each locomotive for the Trip Optimizer system. Power modes are as follows:

- DB Only.
- Running (power and DB cut in).
- Isolated.
- DB Cutout.

To change the running status of a locomotive:

- Use the Down Arrow and Up Arrow to highlight the desired locomotive, then press the appropriate key to change power mode.
- When all changes are complete, press the Accept key to save changes and return to Train Setup.
- The Cancel key discards all changes and returns to Train Setup.

7.2.5 Change Train Data

The Change Train Data screen allows editing of railcars, loads, empties, tons, length, axles, and maximum train speed.

Use the Up and Down arrows to navigate between items in the train data list. Number keys are used to enter new values for any given train data element. When all train data edits are complete, press the Accept key and tons per operative brake will automatically recalculate, if necessary.

Accurate train data is critical to system performance. When changes are entered, ensure all necessary fields are edited, including maximum train speed. Inaccurate information can lead to poor system performance.

7.2.6 Verify Restrictions

Form A and B restriction information will not be displayed in non-Integrated mode. To verify restrictions, perform the following:

- If all restrictions are correct, press the Accept key.
- If restrictions are incorrect, press the Reject key and exit.
- Press Yes key to confirm.
- Press Start Trip key.

7.2.7 Track Selection

Once Initialization is complete, the rolling map will appear and the system will prompt for current track. Select the appropriate track to continue. When a train is initialized on other than main track, the appropriate track may not be listed. Select the main track option that the train will occupy upon entering the main.

7.2.8 Engaging Auto Control

The following requirements apply when engaging auto control:

- Do not comply with Trip Optimizer prompting or utilize auto control when:
 - Any part of the train is on a siding or other-than-main-track location.
 - Movement is within 2 miles of the end of a Track Warrant Authority.
 - Except as provided for in Auto Through Divergence, movement is operating on a signal less favorable than a proceed indication.
- When train speed is above 12 mph, the Auto Control Available message is displayed.

To engage Auto Control, perform the following:

- Press Auto Control key.
- Within 30 seconds, move the throttle handle to notch 8.
- Auto Control Active message will be displayed, and the auto notch box will appear below the throttle box.

7.2.9 Disengaging Auto Control

To disengage Auto Control, perform the following:

- To immediately reinstate manual control, move the throttle to match the Auto Notch position displayed in the Auto Throttle box; or
- Press Manual Control key and move the throttle to match the Auto Notch position during the 30 second countdown.

7.2.10 Auto Through Divergence

When approaching a main track divergence, approximately 1,500 feet before the approach signal, the system will prompt Track Info Needed Ahead. If a diverging route is selected, Auto Active Calculating will appear. Once the calculations are complete, the rolling map and plan will update to accommodate the diverging route. The prompt will then display Auto Control Active.

If the system determines that it cannot complete the diverging route in auto, Take Manual Control Now will appear.

Auto control may be maintained when the train is performing a main-to-main diverging movement. Under these circumstances, Trip Optimizer can remain in auto control on signal indications more favorable than an approach.

7.3 Trip Optimizer – PTC Integrated Mode (TO/PTC-IM)

When equipped, PTC integrated Trip Optimizer enables the energy management system to act upon critical train and operating data received from PTC.

7.3.1 Identifying Trip Optimizer with PTC Integration

To determine if the Trip Optimizer system is integrated with PTC, perform the following:

- Initialize PTC.
- Press the Trip Optimizer key on the locomotive screen.
- Identify one of two initialization messages displayed on the welcome screen:
 - Setup TO PTC Trip – PTC integrated Trip Optimizer is available.
 - New Trip – PTC integrated Trip Optimizer is inactive or unavailable and will operate in stand-alone mode.

7.3.2 Displays

TO/PTC-IM will not display any Trip Optimizer information on the PTC screen. All functions, indications, and displays are located on locomotive display screens.

7.3.3 Initialization

To initialize PTC integrated Trip Optimizer, perform the following:

- Press the Setup TO PTC Trip key.
- Review all data displayed on the Trip Setup Screen.
- If correct, press the Accept Trip Key.
- Verify locomotive power status provided by PTC by pressing the Change Power Mode key:
 - PTC will not provide the Trip Optimizer system with information related to locomotives operating with dynamic brake only or dynamic brakes cut out; this information must be edited manually.
 - Incorrect locomotive power status will result in poor performance and increase the potential for PTC warnings and/or enforcements.
- The system will transition to the Map screen and display Waiting for PTC on the top of the screen.
- Once PTC is in the active state, the Trip Optimizer rolling map will be displayed.

7.3.4 Operation

The following applies to the use of TO/PTC-IM:

- TO/PTC-IM is designated for use on PTC track only.
- Operators are prohibited from complying with Trip Optimizer prompts when any part of the train is located on non-PTC track.
- When PTC is in any state other than active, Trip Optimizer will not offer auto control.
- Movement must reach a minimum speed of 12 mph to activate auto control.
- The minimum speed to maintain auto control is 9 mph.
- TO/PTC-IM will require manual control.
 - Two miles from a PTC generated stop target.
 - Two miles in advance of, within, and one entire train length beyond Form B limits.
- PTC cannot recognize or enforce High Threat Urban Areas for key trains.
 - Utilize manual control in High Threat Urban Areas.
- Auto control may be used on CTC sidings when operating on a signal more favorable than an approach.

CTC or TWC/ABS Territory

- Auto control must be used when operating on any signal more favorable than an approach indication.
- Auto control may be used when performing a main track to main track diverging movement.
- TO/PTC-IM will plan to operate 35 mph beyond approach signals communicated by PTC.
- After passing an approach signal, auto control will not be available until the lead locomotive has passed a signal more favorable than an approach.

TWC Territory

- Auto control must not be used within two miles of the end of the train's authority.

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1.0 Distributed Power System Overview

Distributed power systems are designed to provide synchronous or asynchronous control of remote locomotive consist(s) in a train. Commands are delivered via air brake and/or radio signals.

The locomotive engineer controls each locomotive consist in the distributed power train from the lead locomotive.

The term remote locomotive applies to the controlling locomotive in a remote consist. Locomotives connected by trainline for multiple unit service to the lead and remote locomotives are considered trail locomotives.

2.0 Preparing the Distributed Power System

2.1 Locomotive Conditioning and Testing Order

Condition and test distributed power locomotives in the following order:

- Condition remote locomotive(s).
- Condition the lead locomotive.
- Link to remote locomotive(s) from the lead locomotive.
- Perform a Brake Pipe Test.
- After remote locomotive(s) are placed in the train, perform a Flow Sensor Test.
- Perform a Direction Verification Test.
- Test the emergency feature if a remote is positioned at rear of train and is not equipped with a standalone EOT.

2.2 Conditioning Remote Locomotive(s)

Ensure that any PCS or other air brake faults are reset before attempting set up distributed power. To condition a remote locomotive, perform the following:

- Control Stand:
 - Engine Run – Off.
 - Generator Field – On.
 - Control or Control/Fuel Pump – On.
 - Dynamic Brake – On.
 - Power Limit Switch, if equipped – Enabled.
 - Reverser – Removed.
 - Independent Brake – Lead and fully applied.
 - Automatic Brake – Cut in and released.
- Engine Control Panel:
 - Engine Control / Isolation Switch – Run.
 - LEB Computer – On.
- Circuit Breakers:
 - DP Radio – On.
 - DP Trainline – On (Not present on all locomotives).
- Locomotive Screen:
 - Select Distr Power.
 - Select Remote Setup.

- Enter lead locomotive number.
- Select direction of the remote in relation to the lead locomotive (Same or Opposite).
- Select Accept.
- Verify Lead Cut In and DP Enabled.

After the above steps are complete:

- Place the independent brake valve in release.
- Place the automatic brake handle in Handle Off.
- Insert the keeper pin in the automatic brake handle, if equipped.
- Place trailing headlight on dim if locomotive is placed at the rear of a train.
- Lock the locomotive engineer's seat so it does not swivel.
- Pull in mirrors and close windows.
- Lock the doors.
- Set up all locomotives coupled to the remote in conventional trailing mode.

2.3 Condition the Lead Locomotive

Ensure that any PCS or other air brake faults are reset and equalizing reservoir is adjusted to the required pressure before attempting setup. To condition the lead locomotive, perform the following steps:

- Control Stand:
 - Engine Run – On.
 - Generator Field – Off.
 - Control or Control/Fuel Pump – On.
 - Dynamic Brake – On.
 - Power Limit Switch, if equipped – Enabled.
 - Reverser – Centered.
 - Independent Brake – Lead and fully applied.
 - Automatic Brake – Cut in and released.
- Engine Control Panel:
 - Engine Control / Isolation Switch – Run.
 - LEB Computer – On.
- Circuit Breakers:
 - DP Radio – On.
 - DP Trainline – On (Not present on all locomotives).
- Locomotive screen:
 - Select Distr Power.
 - Select Lead Setup.

The locomotive is now conditioned and ready for the linking process.

2.4 Linking Lead and Remote Locomotive(s)

Perform the following steps to link the lead and remote locomotive(s):

- Enter the locomotive ID and number of the remote locomotive closest to the lead locomotive.
- Press Link.
- Ensure the screen displays Link OK.
 - If Link Fail is displayed, check and re-enter the number of the remote locomotive.
 - If Link Fail is displayed after subsequent attempt, check the setup of the remote locomotive.
- Repeat the above steps for each additional remote locomotive consist.
 - Ensure that remote locomotives are entered in accordance with their physical placement in the train, from front to rear.
- Press Done or Accept.
- Press DP Main Menu and check event and alarm logs for faults or other conditions.
 - Air brake system failures and back-up emergency valve failures must be corrected before departure.
- Press Exit.

2.5 Brake Pipe Test

To perform a Brake Pipe Test, perform the following:

- Ensure the independent brake is fully applied.
- Recover PCS.
- Wait for the message Perform Brake Pipe Test.
- When flow rates have stabilized and sum of air flow is less than 90 CFM and no more than 60 CFM at any location, Press Brk Pipe Test.
- Press Execute.
- When prompted, move the automatic brake valve handle to Minimum Reduction.
 - Moving beyond the Minimum Reduction will void the test.
- If Successful, Brake Pipe Test OK will be displayed.
- If unsuccessful, repeat the above steps.
- Select DP Main Menu.
- Press Mode.
- Press Run (Or Run – FTE).
- Press Execute.
- Press Exit.

2.6 Validation of Locomotive Sequence

Following the set up of the distributed power system, adjustment to locomotive consist(s), or change in locomotive positioning, the locomotive engineer must ensure that remote locomotives are sequenced properly in accordance with physical placement in the train.

2.7 Flow Sensor Test

A Flow Sensor Test validates the Comm Loss Idle Down feature of the distributed power system.

A Flow Sensor Test is required:

- At origin, after locomotives have been conditioned, linked, and placed in the train.
- At other than origin.
 - The first-time cars are placed between a lead locomotive and remote or between remotes.
 - Anytime 60 or more cars are added between a lead locomotive and remote or between adjacent remote consists.

To perform a Flow Sensor Test, comply with the following:

- Release the automatic brake.
- Ensure that air flow complies with AFM Test Requirements.
- Place the Distributed Power Radio circuit breaker to off.
- Make at least a 20-psi reduction with the automatic brake.
- Wait for service exhaust to cease.
- Place the Distributed Power Radio circuit breaker to on.
- Ensure remote(s) display FLOW OUT, which will indicate a successful test.
- From the Remote Menu screen, normalize each remote by pressing Normal, then Execute.
- Move the automatic brake handle to release.
- If the test is unsuccessful, repeat the steps above.

When a lead locomotive is not equipped with breakers necessary to perform the steps listed above, perform a train check.

A Flow Sensor Test will fulfill the requirements of a Class III air brake test when conducted in accordance with required pressures.

2.8 Directional Verification Test

A directional verification test must be performed when:

- A train is first assembled with distributed power locomotive(s).
- It is suspected that the remote is not providing power in the same direction as the lead locomotive.

To perform a directional verification test, complete the following:

- Ensure all distributed power locomotives are in Run mode with the generator field on.
- Place all remotes in Idle, except for the remote to be tested.
- Place the reverser in forward or reverse.
- Select move to back and ensure the fence (divider line) is between Lead and Remote.
- Select Traction and Execute.

- Release the independent brake and increase throttle effort by using the More Traction button.
- Validate that the train or train slack moves in the direction of travel indicated by the reverser.
- Repeat these steps to test each remote consist.
- When complete, select Move to Front, Normal, and Execute.
- Validate that remote(s) return to Normal and display flow rates.

2.9 Remote Locomotive in Lieu EOT

Remote locomotives may be used in lieu of a two-way EOT when a remote consist is placed at the rear of the train.

To test the emergency feature of a locomotive placed at the rear of a train without an EOT, perform the following:

- Ensure that air flow complies with AFM Test requirements.
- To test the emergency feature with a locomotive placed at the rear of a train:
 - Close a brake pipe angle cock anywhere ahead of the remote locomotive on the rear of the train.
 - Trains with multiple remote consists must have an angle cock closed at a location between the last mid remote and the remote locomotive on the rear of the train.
 - Place the automatic brake handle of the lead locomotive in emergency.
 - Verify the emergency brake application propagates as required.
- To test the emergency feature of a light locomotive consist:
 - Close the brake pipe angle cock between the remote that will be placed on the rear of the train and the lead consist.
 - Place the automatic brake handle of the lead locomotive in emergency.
 - Verify the emergency brake application propagates as required.
- Record the test results on the Crew Information Form.

3.0 Distributed Power Operating Modes and Features

3.1 System and Remote Mode Description

System and individual remote modes determine how each remote consist will respond to commands from the lead locomotive.

3.2 System Operating Mode

3.2.1 Changing System Operating Mode

After completing a brake pipe test or brake pipe leakage test, change the system mode to Run to enable traction and dynamic brake functions of the distributed power system.

To change the operating mode, perform the following from the controlling locomotive:

- Select DP Main Menu.
- Select Mode.
- Select Run or Idle.
- Press Execute.

Run or Idle status will be displayed on the screen.

3.3 Remote Operating Modes and Features

3.3.1 Normal

In normal mode, all remote traction, dynamic brake, and air brake functions are enabled. The brake valve may be cut in. When the distributed power system is linked and the brake pipe test is completed, the remote consists are in Normal.

To return a consist to Normal mode:

- Select Normal on the remote display.
 - Locomotive(s) in Stop mode should be placed into Isolate mode to be restarted before returning to Normal mode.
- After changing the remote to Normal, the automatic brake valve may be cut in using the following steps:
 - Make at least a 10-psi reduction with the automatic brake.
 - After the brake pipe exhaust has ceased, move the automatic brake handle to release.
 - With the lead locomotive's automatic brake handle in release, the remote brake valve will cut in when the brake pipe pressure increases at least 4-psi at the remote locomotive.

3.3.2 Idle

In Idle mode, throttle and dynamic brake functions are disabled. All remote air brake functions are enabled, and the brake valve may be cut in.

To place a consist in Idle, select Idle on the remote display. This may be changed while stationary or moving.

3.3.3 BV Out

When in BV Out mode, a locomotive will not respond to propagate air brake signals. A remote can be changed to BV Out while stationary or moving.

To cut in a brake valve on a remote locomotive that has been returned to Normal or Idle:

- Make at least a 10-psi reduction with the automatic brake.
- After the brake pipe exhaust has ceased, move the automatic brake valve handle to release. The brake valve will be cut in when the brake pipe pressure increases at least 4-psi at the remote locomotive.

3.3.4 Isolate

In Isolate mode, the remote's throttle will not respond to commands and the brake valve is cut out. All air brake functions are disabled except emergency air brake and independent brake. A remote consist changes to Isolate when a communication loss idle down has occurred.

3.3.5 Set Out (S/O)

In set out mode, the remote's throttle does not respond to commands and remains in Idle. The remote's emergency brake application is enabled, and the independent brakes are fully applied. All other air brake functions are disabled, and the brake valve is cut out.

Set Out (S/O) mode is used to condition a remote consist left standing uncoupled from the controlling locomotive of a distributed power train.

3.3.6 Stop

In Stop mode, the remote's throttle is set to Engine Stop. All other remote functions are set to Isolate. Use Stop mode to shut down locomotives when an emergency exists, such as impact, fire, hazardous material leak, or stop in a tunnel. The Stop mode will shut down all locomotives in the selected remote consist.

Execute Stop mode as follows:

- From the Remote display, select the remote consist to be shut down.
- Select Stop.
- Select Execute.

To return a remote to Normal after executing Stop Mode:

- Select the remote consist to be restarted.
- Select Normal.
- Select Execute.
- Manually restart each locomotive in the remote consist(s).

3.3.7 Box Car

On certain locomotive types, remote locomotives will transition to Box Car mode when the following conditions are met:

- Remote(s) are in a state of communication loss and an emergency or penalty brake application occurs.
- The penalty timer has expired.

While in box car mode, the following will apply:

- The lead locomotive will charge the brake pipe.
- When the remote brake pipe pressure rises above 60-psi, the independent brake of the remote consist(s) will release.
- The locomotive engineer will be able to control the train with lead consist locomotives only.
- Box car mode will be maintained until communication is restored.

When communication is restored between the lead and remote locomotive(s), a communication loss idle down event will be generated, and the operator screen will show Isolate mode on the affected remotes. The remote can be returned to Normal using the steps for Normal mode.

Box Car mode cannot be selected manually and does not apply to locomotive(s) in Set Out (S/O) mode prior to communication loss.

3.3.8 BP Rise

When equipped, the distributed power system will monitor brake pipe pressure and respond as follows:

- While operating with communication between lead and remote consist(s), if an unknown brake pipe rise occurs while an automatic brake application is in effect, a brake pipe rise alarm will be displayed on the lead locomotive and the system will continue to operate in the commanded state.
- While operating in a state of communication loss between lead and remote consist(s), if an unknown brake pipe rise occurs while an automatic brake application is in effect, remote(s) will transition to a state of comm loss idle down and enter Isolate mode.

- When an un-commanded brake pipe rise occurs while the reverser is centered and the independent brake is fully applied, the system will initiate an emergency brake application.

3.3.9 Remote Sensed Emergency Brake Application

If equipped, remote locomotives that sense an emergency brake application will respond as follows:

- Report PC and zero brake pipe pressure.
- Propagate the emergency application through brake pipe and radio command.

This feature does not apply to remote(s) in Set Out (S/O) mode.

4.0 Modifying or Ending Distributed Power Set Up

4.1 Unlinking a Lead Locomotive from Remote Consist(s) – Ending Distributed Power

When ending distributed power operation, end distributed power on the lead locomotive first.

To unlink a lead locomotive:

- Ensure the train is stopped with the independent brake fully applied and throttle in Idle.
- When distributed power is ended, the system will initiate a penalty application. Secure the equipment to protect against unintended movement, if required.
- From the DP Main Menu:
 - Select System.
 - Select Unlink.
 - Select Execute.
- A penalty brake application will occur.

4.2 Unlinking Remote(s) – Ending Distributed Power

Complete the following steps after ending distributed power on the lead locomotive.

To unlink remote locomotive(s):

- Fully apply the independent brake.
- On the locomotive screen:
 - Select Distr Power.
 - Select End Dist Pwr.
 - Press Execute.
- Ensure DP On or DP Remote Enabled indicator is off.
- Turn off DP circuit breakers.
- Set up the control stand for either lead or trail operation.

4.3 Adding or Removing Locomotive(s) in Lead or Remote Consist While Linked

If it is necessary to change out the controlling locomotive of a remote or lead consist:

- End distributed power.
- Prepare the distributed power system as required by this section.

4.4 Changing Ends

To change ends on a distributed power train:

- End distributed power.
- Prepare the distributed power system as required by this section.

4.5 Incremental Link and Unlink

Reserved.

5.0 Distributed Power Air Brake Tests

5.1 AFM Air Brake Test

To perform an AFM air brake test with a distributed power train:

- Press More Menu (if displayed).
- Press Remote Menu.
- Ensure each remote displays a Flow value.
- If a remote valve is out:
 - Make a 10-psi reduction with the automatic brake.
 - Press Normal.
 - Press Execute.
 - Move the automatic brake handle to release.
 - Remote(s) must see a 4-psi rise in brake pipe pressure to enable a brake valve to cut in.
- Ensure flow does not exceed the values allowed by an AFM test and that air pressure meets the requirements of the air brake test being performed.
- When the signal is given to apply the brakes, make the required reduction with the automatic brake.
- When the signal is given to release the brakes, move the automatic brake to release.

5.2 Brake Pipe Leakage Test

A brake pipe leakage test should only be used when an AFM test cannot be performed. To conduct a Brake Pipe Leakage test with a distributed power train, perform the following:

- Press Remote Menu.
- Ensure remote(s) display a flow value.
- Ensure the train's air flow and air pressure meet the requirement of the test being performed.
- Press Distr Power.
- Press DP main.
- Press System.
- Press Leakage Test.
- Press Execute.
- The distributed power system will initiate a 20-psi application of the automatic brake, allow pressure to stabilize, and automatically measure leakage.
- The test will take up to 3 minutes to complete and display pass or fail on the display screen.
- When prompted, place the automatic brake handle in the full service position.
- When signal is given to release the brakes, move the automatic brake handle to release.

- The remote brake valve(s) will be cut in automatically.
- Ensure each remote displays a flow value and that brake pipe pressure is rising at the rear of the train.
- Ensure the system is in Run mode.

5.3 Train Check

5.3.1 Train Check Requirements

The train check feature verifies that the brake pipe is open and unrestricted between the lead and the remote consist(s).

A train check is required in the following situations:

- Prior to movement anytime the train has been stopped or it is suspected or known that the train's brake pipe has been compromised.

Train checks are not required when:

- Brief stops are made to comply with signal indications or to line switches.
- On ascending or descending grades where train handling techniques require automatic air brakes to remain applied when initiating movement.
- Immediately following a flow sensor test.
- After building, conditioning, and performing an air brake test at the initial terminal, provided the train departs immediately after the brakes are released.

5.3.2 Train Check Procedure

Perform a Train Check as follows:

- Ensure the train is stopped.
- Make the required automatic brake reduction:
 - When a Class III air brake test is not required, make at least a 10-psi reduction.
 - When a Class III air brake test is required, make a 20-psi reduction and validate that brake pipe pressure has decreased at the rear of the train.
- Wait for service exhaust to cease.
- From the System display:
 - Select Train Check.
 - Select Execute.
- Verify that the remote brake valves are cut out by manually checking status or observing screen prompts to perform a brake release.
- When ready to depart, move the automatic brake handle to release.

If the train check is successful, the locomotive screen will show Train Check-OK. If the display shows Train Check Fail, inspect the brake pipe for closed angle cocks, brake pipe blockage, or excessive leakage, and perform an additional train check. If the subsequent attempt fails, contact the proper authority be governed by their instructions. In all circumstances, brake pipe continuity must be validated before the train proceeds.

6.0 Distributed Power Operation

6.1 Synchronous Operation

After a distributed power train has been conditioned and the system has been placed in Run mode, all remote locomotives will operate in the front group and will reproduce the same commands as the lead locomotive. This is considered synchronous operation.

Where possible, operate all remotes in the front group when the train is operating on:

- Near level or gentle undulating grades.
- Continuous descending or ascending grades without severe undulation.

6.2 Asynchronous Operation

Asynchronous operation occurs when a remote consist is controlled independently from the lead locomotive. Asynchronous operation requires the distributed power system to be in Run mode.

The following applies to asynchronous operation:

- Only the consist(s) displayed in the back group will respond to throttle and dynamic brake commands initiated from the summary screen.
- Only the consist(s) operating in the front group will respond to commands made from the lead locomotive control stand.

6.2.1 Beginning Asynchronous Operation

To utilize asynchronous operation, move the remote consist(s) into the back group to provide independent control of the throttle or dynamic braking as follows:

- Press Control Menu (if displayed).
- Press Move to Back.
- On the DP operation screen, verify that a green vertical bar (fence) separates the front and back group.
 - Remotes to the left of the green vertical bar are in the front group.
 - Remotes to the right of the green vertical bar are in the back group.

Important considerations:

- If the remote consists were operating in power or dynamic braking (synchronous) when the rear remote consist was moved to the back group, the rear remote will remain at its present power setting until it is changed using the soft keys on the summary screen.
- If the consists were in Idle when the rear remote consist was moved to the back group, it is necessary to select Throttle/Traction or Dyn/Brake and to execute the command to take independent control of the back group.
- Once executed, the throttle or dynamic brake will move to Position 1.
- The summary screen continues to display both the front and back throttle positions.
- A remote consist moved to a new group will immediately begin changing throttle or dynamic brake positions to match its new group assignment.
- The distributed power system will not allow the back group to operate in dynamic brake if the front group is in traction.

6.2.3 Asynchronous Controls

Utilize the following to control lead and remote consist(s) during asynchronous operation:

- Control remote consist(s) in the back group by pressing more or less Traction/Braking.
- Control the throttle or braking of the locomotive(s) in the front group, with the control stand of the lead locomotive.

6.3 Uncoupling Lead and Remote Locomotive(s) With or Without Cars

To uncouple a lead locomotive from a remote consist that will be left standing:

- Stop the train.
- Fully apply the independent brake.
- Make an automatic brake reduction sufficient to prevent movement.
- From the Remote Menu:
 - Select each remote to be left standing and one at a time.
 - Press BV Out.
 - Press Setout (S/O).
 - Press Execute.
- Ensure each remote reports:
 - Flow Out.
 - Setout.
 - Brake Cylinder 72-psi.
- Advise the crew member when it is okay to close the angle cock on the portion to be moved.
- The standing portion of the train must be left in emergency.
- Verify that each remote being left reports:
 - Brake Pipe 0-psi.
 - Brake Cylinder 45-psi (or higher).
 - PC Alarm (flashing).

6.4 Coupling Lead and Remote Locomotive(s) With or Without Cars

To return remotes to Normal after Recoupling:

- Fully apply the independent brake.
- Make the required automatic brake reduction:
 - On heavy or mountain grades or any grade where locomotive brakes are not sufficient to prevent train movement, make at least a 35-psi reduction with the automatic brake.
 - On grades other than heavy or mountain grade, if train movement can be prevented with the locomotive brakes only, no reduction is required.
- From the Remote display:
 - Select each remote one at a time.
 - Press Normal.
 - Press Execute.
- Slowly open the brake pipe angle cock and allow brake pipe pressure to equalize.
- Move the automatic brake handle to release.
- Ensure each remote displays a value for Flow and PCS alarm indicators extinguish.
- Verify that pressure is rising at the rear of the train.

6.5 Break-in-Two

When a break-in-two occurs on a distributed power train, perform the following:

- Select each remote on at a time.
- Press Setout (S/O).
- Press Execute.
- Ensure each remote reports:
 - Brake Pipe 0-psi.
 - Brake Cylinder 45-psi or higher.
 - PCS alarm.
- Determine which remote(s) are still couple to the lead unit and apply sufficient hand brakes to secure both portion of the train as required.
- Close the angle cock on the lead portion of the train and recover emergency on the lead unit and remote(s) coupled to the lead locomotive.
- If necessary, remove or repair the equipment that cause the emergency application.
- Recouple the train.

6.6 Emergency and Penalty Brake Application Recovery

To recover from an emergency application:

- Secure the train, if necessary.
- For an emergency application:
 - Place the automatic brake handle in emergency.
 - When prompted by the locomotive display, move the automatic brake handle to Handle Off and then Suppression.
- For a penalty application:
 - Place the automatic brake handle in suppression for 8 seconds.
 - Ensure the PCS indicator is extinguished on the lead locomotive.
- Press the More Menu.
- Press Remote Menu.
- Select each remote one at a time.
- Press Normal.
- Press Execute.
- On the DP screen ensure:
 - PCS alarm indicator is extinguished.
 - Each remote displays a value for flow.
 - BP is rising at the rear of the train.
- After the air brake system is recharged, if necessary, a brake pipe reduction sufficient to prevent train movement must be made prior to releasing hand brakes.

6.7 Slow Speed Control

To utilize slow speed control, perform the following:

AC4400 Locomotives

- Place the throttle in Idle.
- Center the reverser.

- Fully apply the independent brake.
- Release the automatic brake.
- On the locomotive screen:
 - Select Distr Power.
 - Select DP Main.
 - Select Mode.
 - Select Speed.
 - Press Execute.
- Adjust the speed using Speed Up and Speed Down keys.
- Place the reverser in forward and throttle in Traction 1.
- Release the independent brake.
- The distributed power system will now control the power to obtain the desired speed automatically; if train moves too quickly, apply up to 10-psi with the independent brake.
- If a stop is requested, place the throttle in Idle.
- To resume, place the throttle in Traction 1.
- To end slow speed:
 - Place the throttle in Idle.
 - Select Idle in the mode screen.

ES44AC Locomotives

- Place the throttle in Idle.
- Center the reverser.
- Fully apply the independent brake.
- Release the automatic brake.
- From the main operating screen:
 - Select Speed Control.
 - Select Slow Speed.
- Adjust the speed using Speed Up and Speed Down keys.
- Press Slow Speed On to active the system.
- Place the reverser in the desired direction, release the independent brake, and place the throttle in Traction 1.
- The distributed power system will now control the power to obtain the desired speed automatically; if train moves too quickly, apply up to 10-psi with the independent brake.
- If a stop is requested, place the throttle in Idle.
- To resume, place the throttle in Traction 1.
- To end slow speed control:
 - Place the throttle in Idle.
 - Select Slow Speed Off from the locomotive screen.

Regardless of locomotive type, the distributed power remote mode will override the Speed Control mode. If a remote was placed in Idle or Isolate, it will remain in this state during speed control operation.

6.8 Tower Control

Tower mode is used at certain facilities to allow tower control of a train. Once tower mode is active, the distributed power system will remain in Idle until the tower takes control.

To activate Tower mode:

- Press DP Main.
- Press Mode.
- Press Tower.
- Press Execute.
- Follow on-screen prompts to place the automatic brake to full service and independent brake to release.
- Remove the reverser.
- Place the Generator Field breaker to off.
- Advise tower personnel that the train has been set to Tower mode.

To deactivate Tower mode:

- Ensure the train has a Full Service or Penalty Brake application in effect.
 - If not, the tower still has control.
- Press DP Main.
- Press Mode.
- Select Idle.
- Select Run.
- Press Execute.
- Charge the air brake system and perform the required air brake test(s).

6.9 Leaving a Train or Locomotive(s) Unattended

To leave a train or locomotives unattended, complete the following:

- Secure the train as required.
- On the locomotive screen:
 - Press Distr Power.
 - Press DP Main.
 - Press Mode.
 - Select Idle.
 - Select Exit.
- Ensure the Engine Control Switch of lead and remote consist(s) are left on Run.
- Condition locomotives as required.

6.10 Remote Independent Application and Release

- When the independent brake on the lead locomotive is moved to the desired position, the command will be sent to the remote locomotives provided it is not in Set Out mode.

7.0 Distributed Power Restrictions

7.1 Distributed Power Locomotives Coupled to Lead Consist

Unless authorized by Operations, distributed power locomotives must not be coupled to the lead consist of a train. When authorized, ensure:

- Dynamic brake does not exceed allowable limits.
- Brake pipe and main reservoir hoses are coupled between the lead and remote consists.

- When all operating locomotives, including remotes, are located on the headend, the train complies with conventional train TrAM requirements.
- All automatic brake valves, except the lead locomotive, are cut out.
- Trip Optimizer is not used.

7.2 Operating Restrictions – Mixed Distributed Power Trains

Except for trains operated by Energy Management or when authorized by local operating instruction, the following requirements apply to distributed power operations with mixed distributed power trains:

- While in asynchronous operation, remote locomotive(s) must use the same or lower throttle position as the lead locomotive consist.
 - Exception: When operating a heavy train on ascending grade, the throttle position of the remote(s) may be higher than the lead locomotive consist. Locomotive engineers must be vigilant to ensure acceptable in-train force levels are maintained.
- When dynamic brake is used on the lead consist, do not use throttle position(s) on remote locomotives.

7.3 Maximum Distances Between Locomotive Consists

If combined air flow to the brake pipe does not exceed 60 CFM, the maximum distance between the lead and remote consist is 10,000 feet.

If combined air flow to the brake pipe is between 61 and 90 CFM and no more than 60 CFM at any one locomotive consist, the maximum distance between any of the locomotive consists is:

- Single Remote Trains: 5500 feet.
- Multiple Remote Trains: 7500 feet.

7.4 Handling Dead or Disabled Remote Locomotives on Rear of Train

In the event a remote locomotive becomes dead or disabled enroute, the train must be conventionalized or remarshalled at the first available location. The train may proceed to an alternate location, provided the following conditions are met:

- Communication is maintained between lead and remote consists.
- Remote locomotives are set to Isolate mode with the remote feed valve cut out.
- Main Reservoir pressure on the remote is 85-psi or greater.
- Train passes a TrAM check.
- The dead or disabled locomotive is drained, if required.

When a remote locomotive's main reservoir pressure is depleted, the locomotive brakes will be inoperative.

If the main reservoir pressure drops below 85-psi or sustained communication loss occurs with the remote, the movement must not exceed 25 MPH and proceed only to the first available location where the train can be conventionalized or locomotives replaced.

8.0 Distributed Power Communication

8.1 Distributed Power Communication Check

The distributed power system checks radio and onboard computer communication of the lead and remote locomotives. If a communication check fails, the system will:

- Attempt to operate the lead and remote backup radios.
- Declare a communication interruption at the lead and at the remote consist.

8.2 Lead Locomotive Communication Interruption

If a lead locomotive fails to receive an expected command reply from a remote locomotive, a yellow COMM light will illuminate or COMM light will flash.

The lead unit will declare a communication interruption:

- If no reply is received from a remote unit for 45 seconds; or
- In 10 seconds, following an automatic brake pipe reduction.

When a communication interruption is declared, a red COMM light will appear and a double chime will sound.

The status displayed for the remote consists will not change during the COMM interruption and will continue to indicate the remote status at the time of the failure.

8.3 Remote Locomotive Communication Interruption

If communication loss with remote locomotives occurs, each remote consist will:

- Continue to operate on the last throttle and air brake command received via the radio.
- If no change in brake pipe pressure is detected at the remote, the remote locomotive(s) will:
 - Continue to act on the last command received for up to 90 minutes.
 - At the end of the 90-minute override, initiate a penalty brake application if radio communication has not been reestablished.

The remote brake valve must be cut in and charging at the time of and during the communication interruption to allow the remote consist to operate in override. If a remote brake valve is cut out, remote will immediately go to idle in the event of a COMM interruption. Certain locomotives will retain dynamic braking effort regardless of brake valve condition. Locomotive engineers must monitor gauges to assess dynamic brake status. Remote consists must be returned to normal after COMM has been reestablished.

If possible, during a communications interruption between the lead and remote consists, keep the train moving to a location where communications might improve. If distributed power is needed, avoid making an automatic brake application.

8.4 Comm Loss Idle Down

To drop out a remote brake valve in a state of communication loss, make a straight away automatic brake reduction of at least 20-psi. When service exhaust stops, the remote brake valve will be out and automatic override nullified.

Comm Loss Idle down requires a significant change in brake pipe flow rates to take effect. An emergency brake application may be required while operating with comm loss when:

- The automatic brake was released immediately before loss of communication.
- The train has significant brake pipe leakage.
- A 20-psi brake pipe reduction was already in effect.

8.5 Radio Communication Re-established

When radio communication is reestablished, at the lead locomotive:

- Return the remote to Normal mode to restore throttle and dynamic brake functions.
- Cut in the automatic brake valve.

9.0 Alarms and Displays

9.1 Audible Alarms

The control console includes an audible alarm which sounds to alert the locomotive engineer of certain conditions. Audible alarms are as follows:

- Single Chime – Information only, such as brake pipe test completed.
- Double Chime – Alarm or conflicting condition, such as brake warning.
- Triple Chime – Major alarm such as sustained wheel slip or locked axles.

9.2 Alarm Displays

The following alarms may be displayed on the Distributed Power screen:

- MU Alarm (White) – General locomotive alarm, example hot engine.
- Alarm – New data is on the Alarm screen.
- Comm (Yellow) – Communication interrupted for less than 45 seconds.
- Comm (Red) – Communication interrupted for more than 45 seconds.
- Sys Fail (Red) – Major system malfunction; emergency brake is applied and further system operation is not allowed.
- Whl Slip (Red) – Continuous wheel slip condition; illuminates momentarily to indicate non-continuous slip.
- PCS (Red) – PCS open at lead or remote locomotive.
- Locked Axle (Red) – Indicates a locked axle condition.
- Eng S/D – Engine has shut down.
- Auto TMC – Automatic traction motor cut out.

Section 8 – Head of Train (HOT) and End of Train (EOT) Systems

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1.0 HOT-EOT System Setup

1.1 Installation

- The calibration date shown on the EOT device must not be expired.
- Install the EOT
- From the EOT display, verify that the EOT battery voltage is 12.2V or higher.
- Enter the EOT number in the HOT on the lead locomotive.
- Press the HOT *COMM TEST* button and verify on the HOT display that the Comm Test passed.
- After charging the train, validate that the pressure displayed on the HOT is within +/- 3 psi of the EOT pressure.

1.2 Repeating

- On trains equipped with a mid-train train distributed power locomotive, configure the HOT repeater as follows:
 - DPS Device – Ensure the HOT is disarmed and the EOT ID is set to zero.
 - Integrated LXA – Enter the EOT ID into the HOT.
 - Quantum Q3442 – Enter the EOT ID and place the toggle switch in the repeater position.

2.0 Arming the HOT

To arm the HOT:

- Press the button on the EOT.
- When the Arming message appears on the HOT, immediately press the Arm button on the HOT.
- Verify that the HOT indicates Armed or Emergency Enabled.
- If the system fails to Arm, press the HOT COMM TEST button to determine communication status and repeat the steps above.

3.0 Testing the HOT and EOT System

An HOT/EOT system must be tested at the initial terminal or other point of installation to determine that the device can initiate an emergency brake application. To test the system, perform the following:

- Close the angle cock ahead of the last car.
- Activate the EOT emergency toggle switch in the locomotive cab.
- Verify that the cars behind the angle cock go into emergency.
- Allow 15 seconds for the EOT emergency valve to close before opening the angle cock.
- Open the angle cock.

Alternative testing methods:

- Test the system with the EOT mounted on the locomotive. Perform a comm test after installation on train.
- Individual components are bench tested by Mechanical. Perform a comm test after installation on train.

4.0 Disarming the HOT

To disarm the HOT:

- Initiate the Disarm function on the HOT and set the EOT ID code to “00000”.
- Verify the HOT displays “disarmed” or “emergency disabled” and an EOT ID of 00000.

Disarm the HOT when:

- A train reaches its final terminal.
- Replacing an HOT or EOT enroute.

5.0 Emergency Toggle Switch

The HOT/EOT system must be properly armed to activate an emergency brake application. To initiate an EOT emergency brake activation:

- Lift the red cover and toggle of the EOT emergency switch.
- Verify the rear brake pipe pressure reduces to 0 psi.

When an EOT emergency brake activation is initiated, the HOT will continually transmit a series of emergency brake commands to the EOT for two minutes or until the EOT acknowledges the command. If, after two minutes, the EOT has not responded and the rear brake pipe pressure is not 0 PSI, the locomotive engineer will be required to reinitiate the EOT emergency brake activation.

6.0 Emergency Application Capability from Rear of Train

All trains must have the ability to initiate an emergency brake application from the rear of the train, except:

- Locomotives operating without cars.
- Passenger trains.
- Locals, road switchers, work trains, and yard assignments provided:
 - Train weight does not exceed 4,000 trailing tons.
 - Movement does not operate over heavy or mountain grade or travel a distance farther than can normally be operated by a single crew in a single tour of duty.
- Trains that operate at speeds less than 30 mph and do not operate over heavy or mountain grades.

7.0 Acceptable Methods for Initiating Emergency Application at Rear of Train

The following methods will fulfil the requirement to initiate an emergency application from the rear of the train:

- An operable HOT and EOT which has been armed and tested at the point of installation.
- A Distributed Power locomotive at the rear of the train which is linked to the lead locomotive.
- Helper locomotive(s), caboose, or passenger equipment at the rear of the train provided it is equipped with:
 - An emergency brake valve.
 - An employee with a two-way voice radio in communication with the locomotive engineer in the lead locomotive.

8.0 HOT-EOT System Enroute Failure (or consider calling this “Loss of Emergency Application Capability from Rear of Train”)

8.1 Requirements

Trains required to be equipped with rear of train emergency capability are considered to have an enroute failure when one of the following conditions occur:

- HOT indicates Front to Rear communication loss. Rear to Front communication loss is NOT considered an enroute failure.
- EOT is not Armed or has lost Armed status on the lead locomotive.
- HOT indicates EOT emergency valve failure.
- HOT indicates EOT dead battery.
- HOT indicates Emergency Disabled
- Distributed power communication loss exceeding 16 ½ minutes.
- A loss of voice radio communication between helper locomotive(s), caboose, or passenger equipment at the rear of the train and the lead controlling locomotive.

8.2 Responding to an Enroute Failure (Loss of Emergency Application Capability)

When an enroute failure occurs, trains must not:

- Exceed 30 mph until the failure is corrected.
- Pick up cars enroute.
- Descend or move from a stop on heavy or mountain grade as listed in the timetable, unless the train is equipped with an alternative method to initiate an emergency brake application from the rear of the train. Alternatives include helper locomotives, occupied cabooses, or radio control locomotives. All movements must be made under the direction of local management.

9.0 Loss of Communication

The EOT/HOT system utilizes the following parameters for communication loss:

- Front to Rear – communication loss occurs when the HOT does not receive a response to its Comm Test messages from the EOT for 16 ½ minutes.
- Rear to Front – communication loss occurs when an HOT has not received communication from the EOT for 5 minutes.

Additional information:

- HOT/EOT communication may be restored by moving out of a communication loss area.
- Rear to Front communication loss alarm will clear when the HOT receives any message from the EOT.
- Front to Rear communication loss alarms will clear when the HOT receives a response from the EOT to a COMM Test message.

10.0 Handling EOT Devices

EOT devices may be transported as follows:

- In locomotive EOT holders.
- Installed on the coupler of a distributed power locomotive at the rear of a train.
- In the locomotive's engine room or other designated locomotive tool compartment, provided the EOT device(s) do not create a tripping or other hazard and movement is authorized by management.
- Ensure EOT equipment is handled with care to avoid damage to antennas and other internal components.

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1.0 TrAM Areas

Train Area Marshalling areas are:

- Defined by a combination of grade and curvature.
- Divided into six areas.
- Identified in the subdivision page of the timetable.

2.0 Train Types

All trains are classified as one of the following:

- Heavy Bulk:
 - Contains only conventional cars.
 - All cars on the train weigh at least 100 tons and measure 65 feet or less in length.
- Light Bulk:
 - Contains only conventional cars.
 - All cars on the train weight less than 50 tons and measure 65 feet or less in length.
- Heavy Uniform:
 - Contains either conventional or multi-platform cars.
 - All cars on the train:
 - Weigh at least 50 tons per car or platform.
 - The maximum weight difference between cars or platforms does not exceed 20 tons.
 - Maximum length difference between cars or platforms is ten feet.
- Mixed:
 - Any train that does not qualify as one of the train types listed above.

4.0 Train Configurations

4.1 Conventional Train

- A conventional train is a train in which all operating locomotives are located at the head end of a train.
- Conventional trains that meet the definition of Heavy Bulk, Light Bulk, or Heavy Uniform must not exceed 10,000 feet. Mixed conventional trains must not exceed 10,000 feet or 10,000 tons.
 - Exception: Unless otherwise specified, trains with solid intermodal equipment may operate to a maximum length of 12,000 feet.

4.2 Distributed Power Train

- A distributed power train is a train in which all operating locomotives are located at the head end of the train and in up to three additional positions throughout the train. All locomotives are controlled from the head end locomotive consist.
- The maximum length of a distributed power train must not exceed 14,000 feet.

5.0 TrAM Messages

TrAM messages contain the following parts:

- Part 1 – Train Information: provides essential information about the train, including train type.
- Part 2 – Caution Messages: Provides warnings, information, and instructions.
- Part 3 – Marshalling Violations: Lists marshalling violations that require correction before a train can proceed.
 - “All Areas” messages apply to all trains.
 - TrAM Area 1-6 messages apply only to specific TrAM areas.

Example:

PART 3 – MARSHALINGVIOLATIONS

***** ALL AREAS *****

NO VIOLATIONS THAT APPLY TO ALL AREAS

PASSES REMOTE ZONE MARSHALLING

CD RULE 1 - NO CUSHIONED DRAWBAR RESTRICTIONS

***** AREA 1 *****

NO SPECIFIC VIOLATIONS FOR THIS AREA

***** AREA 2 *****

NO SPECIFIC VIOLATIONS FOR THIS AREA

***** AREA 3 *****

NO SPECIFIC VIOLATIONS FOR THIS AREA

***** AREA 4 *****

MAXIMUM TRAILING CAR TONNAGE EXCEEDED ON 02 CARS

ETTX908301 ETTX803330

***** AREA 5 *****

MAXIMUM TRAILING CAR TONNAGE EXCEEDED ON 02 CARS

ETTX908301 ETTX803330

***** END OF TRAIN AREA MARSHALLING MESSAGES

6.0 Verifying Train Area Marshalling

Supervisors in major yards and terminals are responsible for compliance with TrAM rules. Train crews are responsible to ensure that there are no marshalling violations applicable to the TrAM area(s) they will operate on. All violations must be promptly reported to a supervisor and train dispatcher.

Check the following:

- Train consist:
 - Consist header messages.
 - Car information messages.
 - Train statistics messages.
 - Dangerous commodity messages.
 - Part 1, 2, and 3 Train Area Marshalling messages.
- Crew Information form for work performed enroute.

If there are no marshalling violations the train may proceed complying with all caution messages.

7.0 TrAM Check Requirements for Work Performed Enroute

7.1 Conventional Trains

The following will govern TrAM check requirements for conventional trains performing work enroute.

Heavy Bulk Trains:

- Not required for setout or pickup of operating locomotive(s).
- Not required for setout of cars.
- Not required for pick up of cars that meet the criteria of a Heavy Bulk train.

Light Bulk:

- Not required for setout or pickup of operating locomotive(s).
- Not required for setout of cars.
- Not required for pickup of cars that meet the criteria of a Light Bulk train.

Heavy Uniform:

- Not required for setout or pickup of operating locomotive(s).
- Not required for setout of cars.
- Requirements for a Mixed Conventional train apply when picking up cars.

7.2 Distributed Power Trains

Except for Heavy Bulk trains originating at a customer facility, distributed power trains must always receive a TrAM check after equipment is picked up or set out of the train.

8.0 Documenting Pickups and Setouts

The following requirements apply after performing work enroute:

Planned pickup or setout:

- Leave updated documents with the train.

Unplanned pick up or setout:

- Record the following on the Crew Information form:
 - Work performed.
 - TrAM status (complete – pass or fail, or incomplete – required) and AGWZ compliance.

Locomotive:

- Record the following on the Crew Information Form:
 - Pickups.
 - Setouts.
 - Isolation of lead or remote locomotives.
 - Cut out traction motors.
 - TrAM status (complete – pass or fail, or incomplete – required) and AGWZ compliance.

9.0 Enroute TrAM Violations

When a TrAM marshalling violation is discovered enroute:

- Stop the train.
- Contact the dispatcher.
- Comply with the dispatcher's instructions to correct the marshalling violation.

If authority is granted to move a train with a marshalling violation, it can only be operated to the first location where the violation can be corrected. Train crews must comply with any restrictions assigned to the movement.

10.0 Cushioned Drawbar Rules

10.1 Overview of Cushioned Drawbar Rules

Cushioned drawbar rules apply in all TrAM areas. All trains are subject to the three rules below:

- CD Rule 1:
 - No cushioned drawbar restrictions, train may proceed.
 - TrAM Message:
 - CD RULE 1 - NO CUSHIONED DRAWBAR RESTRICTIONS
- CD Rule 2:
 - Train speed is restricted in areas listed in the timetable.
 - TrAM Message:
 - CD RULE 2 - CUSHIONED DRAWBAR SPEED RESTRICTIONS APPLY

- CD Rule 3:
 - Train must be remarshalled or reduced.
 - TrAM messages:
 - CD RULE 3 - CUSHIONED DRAWBAR CARS EXCEED MAXIMUM – REDUCE
 - CD RULE 3 - INSUFFICIENT CD CARS NEAR REAR OF TRAIN – REMARSHALL
 - CD RULE 3 - REMOTE LOCO INCORRECTLY PLACED FOR OVER 40 CD CARS

10.2 Application of Cushioned Drawbar Rules

Cushioned drawbar rules vary based on the number of cushioned drawbar cars in a train, configuration, and train weight. CD Rule notations will be identified in marshalling messages.

10.2.1 Conventional Trains

The following rules apply to conventional trains:

- 0 to 30 cushioned drawbars:
 - Any weight:
 - CD Rule 1 applies.
 - TrAM check not required.
- 31 to 80 cushioned drawbars:
 - 5,000 tons or less:
 - CD Rule 1 applies.
 - TrAM check not required.
 - Over 5,000 tons:
 - CD Rule 1 or CD Rule 2 applies.
 - Indicated on train list.
 - If train is in this category and train consist or TrAM check is unavailable, CD rule 2 will apply.
- 81 to 120 cushioned drawbars:
 - Any weight:
 - Either CD Rule 2 or CD Rule 3 applies.
 - Indicated on train list.
 - TrAM check is required.
 - If all cars are equipped with cushioned drawbars, CD Rule applies and TrAM check for cushioned drawbars is not required.
- Over 120 cushioned drawbars:
 - CD Rule 3 applies.

10.2.2 Distributed Power Trains

The following rules apply to distributed power trains:

- 0 to 40 cushioned drawbars:
 - Any weight:
 - CD Rule 1 applies.
- 41 to 80 cushioned drawbars:
 - 6,000 tons or less:
 - Either CD Rule 1 or CD rule 3 applies.
 - Over 6,000 tons:
 - CD Rule 1, CD Rule 2, or CD Rule 3 may apply.
- 81 to 120 cushioned drawbars:
 - Any weight:
 - Either CD Rule 2 or CD Rule 3 will apply.
- Over 120 cushioned drawbars:
 - CD Rule 3 applies.

11.0 Mixed Train Instructions

11.1 Application

These instructions apply to the following train types:

- Mixed Conventional.
- Mixed Distributed Power.

11.2 Short Car Coupled to Long Car (32/65 and 41/80 Rule)

Requirements

- A car with an outside length less than 32 feet must not be coupled to a car or platform greater than 65 feet in outside length.
- A car with an outside length less than 41 feet (other than operating cabooses or “crew transportation cars”) must not be coupled to a car or platform greater than 80 feet in outside length.

When either of these rules is violated, a message appears in the “All Areas” portion of Part 3 – Marshaling Violations of the train consist.

Exception: Cranes coupled to Idler cars are exempt from this item, and any short car/long car violation message in Part 3 related to the crane/idler combination does not apply.

11.3 Maximum Trailing Car Tonnage for Cars Greater than 65 Feet in Outside Length

Maximum trailing car tonnages apply to all cars that exceed 65 feet in outside length and vary by TrAM area, car type, and weight. A train may not operate over a TrAM area with a Maximum Trailing Car Tonnage violation.

11.4 Threshold Tonnage

Threshold tonnages vary by TrAM area and apply to conventional trains only. When train tonnage, excluding operating locomotives, exceeds the threshold tonnage, Maximum Trailing Car Tonnage and Ascending Grade Weight Zone rules will apply.

Threshold Tonnage by TrAM Area

- Area 1: 9,550 tons.
- Area 2: 8,000 tons.
- Area 3: 4,200 tons.
- Area 4: 6,000 tons.
- Area 5: 2,800 tons.
- Area 6: TBD.

12.0 Maximum Distance between Lead Locomotive and Last Remote Consist

The distance between the lead locomotive and last remote consist must not exceed 14,000 feet. Distributed power trains operating with a remote more than 10,000 feet from the lead locomotive must operate in a multiple remote configuration or move a remote to a mid-train position.



13.0 Dynamic Brake

TrAM will determine the maximum allowable dynamic brake for on distributed power trains. Restrictions listed will be based on thousands of pounds per the effort indicator on the locomotive display.

One of the following messages will be displayed:

- DYNAMIC BRAKE NOT RESTRICTED
- DYNAMIC BRAKE RESTRICTED – DO NOT EXCEED nnn KLBS RETARDING FORCE

14.0 Remote Zones

On Mixed distributed power trains, the cars immediately ahead of the remote locomotive consist(s) will form remote zones. TrAM will analyze the equipment ahead of each remote location and display one of the following:

- The train passes remote zone marshaling.
- Specific cars have failed remote zone marshaling.
- Remote zone rules do not apply.

Remote zone rules do not apply to a Mixed Distributed Power train when the following conditions are met:

- All cars on the train, except cars on the extreme head end, meet the conditions of a Light Bulk train.
 - The "extreme head end" in this case includes all cars up to and including the car furthest from the head end that does not meet the conditions of a Light Bulk train.
- The number of cars or platforms on the extreme head end does not exceed 10 percent of the total number of cars or platforms between the lead and first remote locomotive consist.

Light Bulk trains that require buffer car(s) will generate failed remote zone marshalling messages. Provided a maximum of two loaded buffer cars are positioned on either side of the locomotive, the remote zone marshalling violation can be disregarded.

If a rear remote consist is removed, a maximum of two loaded buffers may be left on the rear of the train.

15.0 Buff and Draft Forces on Distributed Power Trains

When TrAM determines that draft or buff forces are exceeded, TrAM will display one of the following messages:

- DRAFT FORCES EXCEED MAXIMUM LEAD REMOTE 1 REMOTE 2 REMOTE 3
- BUFF FORCES EXCEED MAXIMUM REMOTE 1 REMOTE 2 REMOTE 3

16.0 Ascending Grade Weight Zone (AGWZ) Rules

AGWZ rules apply to TrAM areas 4 and 5. The combination of ascending grade and curvature can create undesirable in-train forces that affect lighter weight cars under certain operating conditions. AGWZ rules do not apply to Heavy Bulk trains.

16.1 Conventional Trains

The following AGWZ requirements apply to conventional trains:

- Light Bulk:
 - Less than 24 driving axles:
 - AGWZ restrictions do not apply.
 - More than 24 driving axles:
 - Weight does not exceed TrAM area Threshold Tonnage:
 - AGWZ restrictions do not apply.
 - Weight is greater than TrAM area Threshold Tonnage:
 - Prohibited.
- Mixed and Heavy Uniform:
 - Less than 24 driving axles:
 - AGWZ restrictions do not apply.
 - More than 24 driving axles:
 - Weight does not exceed TrAM area Threshold Tonnage:
 - AGWZ restrictions do not apply.
 - Weight is greater than TrAM area Threshold Tonnage:
 - Cars or Platforms on head end of train must pass the minimum requirements listed in Conventional Train Head End Weight Requirements.

Train crews are responsible to ensure that Heavy Uniform conventional trains do not operate with AGWZ rule violations. TrAM will not generate messages for this train type.

16.2 Distributed Power Trains

The following AGWZ requirements apply to distributed power trains:

- If TrAM does not display messages indicating that Draft or Buff force have been exceeded, the train complies with AGWZ rules.

16.3 Conventional Train Head End Weight Requirements

Cars or platforms on the head end of Mixed Conventional trains affected by AGWZ rules are restricted as follows:

- TrAM Area 4:
 - First 10 cars or platforms must each weigh a minimum of 60 tons.
- TrAM area 5:
 - First 12 cars or platforms must each weigh a minimum of 75 tons; or
 - First 15 cars or platforms must each weigh a minimum of 60 tons.

When an articulated or solid drawbar connected car extends beyond the first 10, 12, or 15 cars or platforms required by this rule, the minimum weight rules apply to each platform on the car.

17.0 TrAM Messages and Crew Responsibility

Train Area Marshaling Messages for Areas 4 and 5 will indicate, for Mixed Conventional trains only, if the cars or platforms on the extreme head end of the train meet the minimum weight requirements for operation with over 24 equivalent driving axles in an Ascending Weight Zone.

Train Area Marshaling messages will indicate if the cars on the head end of the train meet the minimum requirements for operation with more than 24 equivalent driving axles in Ascending Grade Weight Zones in TrAM Areas 4 and 5.

When operating in an Ascending Grade Weight Zone, crews are responsible to ensure that the number of equivalent driving axles of their train does not cause a violation of the Ascending Grade Weight Zone Rules. TrAM messages do NOT state if the train is operating with too many equivalent driving axles for that train configuration in an Ascending Grade Weight Zone.

Section 10 – Certification

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1.0 Responsibilities of Certified Employees

Locomotive engineers, remote control operators, and conductors must:

- Comply with all certification requirements.
- Ensure driving record and vision and hearing information is submitted within required timelines.
- Know the expiration date of their certification.
- Notify local management no less than 14 days prior to expiration of their certification.
- Notify crew management within 48 hours of expiration of their certification.
- Not work or accept a call to work a certified position beyond their certification expiration date.
- Carry a current certification card or temporary certificate while on duty.
- Present their certification card to the proper authority upon request.
- Be qualified on the territory over which they will operate.
- Complete a territory exam for each territory over which a qualification will be held:
 - During each recertification cycle.
 - When first qualifying over a territory.
 - Before operating over a territory for which a qualification has lapsed.

2.0 Motor Vehicle Driving Records

Locomotive engineers, remote control operators, and conductors must report any convictions for:

- Operating a motor vehicle while under the influence of or impaired by a controlled substance.
- Refusal to undergo such testing when a law enforcement official seeks to find out whether a person is operating under the influence of alcohol or a controlled substance.

State sponsored diversion programs, guilty pleas, and completed state actions to cancel, revoke, suspend, or deny a driver's license are considered as applied to this rule.

An employee must report any conviction to an employee assistance representative and the Manager Operating Practices within 48 hours after the employee receives notice of a conviction. Managers notified of a conviction by an employee must also report this information to the Operating Practices.

3.0 Familiarization

The following general requirements will apply to territory familiarization:

- Federal submissions for CFR 240 and CFR 242 will govern familiarization standards.
- A locomotive engineer not qualified on a specific territory will be provided a minimum of one round trip with a locomotive engineer pilot.
- A conductor not qualified on a specific territory will be provided a minimum of one round trip with a qualified pilot.
- A locomotive engineer or conductor who has not operated over a territory for a period of twelve consecutive months must make a minimum of one familiarization trip.

Glossary

Accelerometer: An instrument that displays in mph per minute the rate of speed increase or decrease.

AC Locomotive: Locomotives equipped with AC traction motors that are not affected by short time ratings.

Actuate: Action required to prevent an automatic brake application on the locomotive(s) or to release locomotive brakes from an automatic brake application.

Air Brake: A system of devices operated by compressed air arranged in a system and controlled manually, electronically, or pneumatically, that retards or arrests the motion of a car or locomotive.

Air Brake System: A collection of devices required for the operating air brakes to control the speed of and stop a locomotive or train.

Air Compressor: A locomotive device, powered by the diesel engine or an electric motor, that compresses air for operating the air brakes and all other air-operated devices on locomotives and cars.

Air Flow Indicator: An instrument that measures the air flow through the automatic brake valve to the brake pipe.

Air Gauge: An instrument that indicates air pressure in pounds per square inch (PSI).

Alerter: A device that initiates a penalty brake application when the locomotive engineer does not maintain the required frequency of movements of the controls.

Angle Cock: A device used to open and close the brake pipe on the ends of locomotives and cars.

Articulated Car: A car constructed by permanently connecting two or more platforms with an articulated joint.

Automatic Brake Valve: A device that controls the flow of main reservoir air into the equalizing reservoir and brake pipe to charge and release brakes.

Automatic Brake Valve Cut Out Valve: A device used to cut the automatic brake in or out.

“B” End: The end of a freight car where the hand brake is located, unless otherwise identified.

Bleed: A term used to describe the venting of air to atmosphere on a car by manual operation of the release valve.

Block of Cars: Two or more cars coupled together that have previously received a Class I or Class IA air brake test for which the Train Brake Status is available.

Block Swap: Adding a maximum of one car or solid block of cars, qualified by the required air brake test and safety inspection, to another through train.

Brake Application: A brake pipe pressure reduction that causes the control valve to move to the service or emergency position.

Brake Cylinder: A metallic cylinder containing a piston. Compressed air forces the piston outward to apply the brakes. When the air pressure is released, the piston returns to its normal position by a release spring coiled around the piston rod inside the cylinder.

Brake Cylinder Release Valve: The valve that initially bleeds brake cylinder pressure. If held open, it will bleed auxiliary and emergency reservoir pressures.

Brake Indicator: A device which indicates the brake application range and indicates whether brakes are applied and released.

Brake Pipe: The pipe, angle cocks, hoses, and hose couplings that distribute compressed air throughout the automatic brake system.

Brake Pipe Gradient: The difference in brake pipe pressure between the locomotive or air source and the rear car of the train.

Branch Pipe Cut out Cock: A device on locomotives and cars that isolates the control valve from the brake pipe.

Brake Pipe Pressure: The amount of pressure contained in the brake pipe, commonly expressed in pounds per square inch (psi).

Buff Force: Compressive coupler forces that occur during a slack-bunched condition.

Communication Loss Idle Down: A distributed power feature that detects a change in brake pipe pressure during an interruption of distributed power radio communication. Also referred to as Comm Loss Idle Down.

Continuity: The ability to transmit a brake signal between the lead locomotive and the last piece of equipment on the train.

Control Valve: The valve on locomotives and cars that charges reservoirs and applies or releases brake cylinder pressure in response to changes in brake pipe pressure.

Coupler Alignment Control: Specially equipped couplers, installed on most locomotives that only allow the coupler in buff to move laterally within certain limits.

Cut Out Cock or Valve: A device used to cut in or cut out certain air brake functions.

Cycle Braking: The process of applying, releasing, and reapplying train brakes to control speed without completely recharging the train brake system.

DC Locomotive: Locomotives equipped with DC traction motors.

Dead Engine Cut Out: A device near the locomotive control valve used when a locomotive unit is handled dead-in-train. When the dead engine cut out cock is opened, the number 2 main reservoir is charged from the brake pipe to operate the locomotive brakes.

Dead-in-Consist: A method of transporting a locomotive unit or units with the diesel engine not running but coupled to an operating unit.

Dead-in-Tow: Hauling locomotive(s) in a train with the brake pipe coupled and the MU air hoses uncoupled.

Distributed Power: One or more locomotive consists that are remotely controlled from the lead, locomotive.

Draft Force: Pulling force, or tension on couplers and draft gear during a slack-stretched condition.

Draft Gear: A cushioning device that transmits draft or buff forces between the coupler and the car or locomotive frame.

Driving Axle: A locomotive axle used for tractive effort.

Dynamic Brake: The system that electronically converts energy created by the momentum of a moving locomotive into an effective retarding brake.

Dynamic Brake Factor: A numerical value used to convey the dynamic braking capability of a locomotive.

Dynamic Brake Holding Feature: A feature of the lead controlling locomotive that allows dynamic braking effort when a PCS open condition exists.

Emergency Application: A reduction in brake pipe pressure at a rate sufficient to cause control valves to move to the emergency position.

Emergency Brake Valve: A valve used to initiate an emergency brake application.

EOT (End of Train Device): A system of components that determines the rear car brake pipe pressure and transmits that information to the display on the head-of-train telemetry device (HOT).

Equalizing Reservoir: A small reference volume on locomotives used to make service brake applications.

False Gradient: A term that describes the amount of pressure present in the rear car while the air brake system is still recharging. False Gradient is any pressure less than True Gradient. The amount of False Gradient can be determined by subtracting False Gradient from True Gradient.

Example:

88-psi – Highest obtainable rear car pressure (True Gradient)

85-psi – current rear car pressure while recharging (False Gradient)

3-psi = Amount of false gradient

Foreign Locomotive: Any locomotive not owned by CPKC.

Foundation Brake Gear: The levers, rods, brake beams, etc. that connect the brake cylinder piston rod to the brake shoes. When air pressure forces the piston out, the brake shoes are forced against the wheels.

Full Service Application: A service reduction of brake pipe pressure that equalizes the pressure between the brake cylinder and the auxiliary reservoir.

Grade: The slope of railway track. Grades are classified as follows:

- Light Grade: any grade below 1.0 percent
- Heavy Grade: any grade between 1.0 and 1.8 percent
- Mountain Grade: any grade greater than 1.8 percent
- Ascending Grade: A section of track where elevation increases
- Cresting Grade: A long ascending grade which rapidly changes to a long descending grade, both of significant magnitude, usually on heavy grade
- Descending Grade: A section of track where elevation decreases

Hand Brake: An assortment of levers, chains, rods, and gears that, when applied by wheel or lever, forces the brake shoes against the braking surfaces to control car or locomotive movement.

HOT (Head of Train Device): A telemetry transmitter/receiver that provides the locomotive engineer with the displays and controls needed to monitor and operate the associated end-of-train telemetry device (EOT).

Helper Locomotive: A locomotive controlled by a helper locomotive engineer used to assist another train.

Horsepower per Ton (HPT): The total horsepower of all working locomotives divided by the total trailing weight of the train in tons.

Independent Brake: A device used to operate the air brakes on a locomotive independent of the automatic brakes.

Initial Terminal: The location where a train is originally assembled.

In-Train Forces: Any force occurring within a train. Examples include buff, draft, run-in, and run out.

Isolation Switch: A multiple position switch used to change the operative status of a locomotive.

Jackknifing: When the center sill between two rail vehicles is excessively misaligned or the coupler is angled sharply. Usually caused by excessive buff forces.

Linking: The process of electronically connecting a lead locomotive to a remote locomotive on a distributed power train.

Light Locomotive(s): One or more locomotives not coupled to cars.

Load Current Meter: An instrument that indicates the amount of current, in amps, flowing through a DC locomotive's traction motor. Also called an ammeter.

Local: A train assigned to perform switching enroute which operates with 4,000 trailing tons or less and travels between a point of origin and a point of final destination, for a distance that is no greater than that which can normally be operated by a single crew in a single tour of duty.

Locomotive Consist: One or more locomotives with multiple unit (MU) connectors coupled, cut in, and operated from a single point of control.

Main Reservoir: A reservoir on a locomotive that stores, cools, and dries the supply of compressed air.

Minimum Reduction: A 7-psi automatic brake pipe reduction.

Multiple Unit (MU) Connections: The hoses and angle cocks at both ends of locomotives that connect air brake equipment in a locomotive consist, allowing operation from a single control. MU connections include the Brake Pipe, Main Reservoir, Actuating, Independent Application and Release, and electrical jumper cable.

Off Air: Not connected to a continuous source of compressed air of at least 60-psi.

Overcharge: When brake equipment on cars or locomotives is charged to a pressure higher than the regulating valve setting.

Penalty Brake Application: An automatic brake application at a service rate caused by a safety control device, such as an electronic alertness device, overspeed, or foot pedal.

Pneumatic Control Switch (PCS): An air operated switch activated by an emergency or penalty brake application that drops the engine speed to idle.

Pressure Maintaining Feature: A system designed to overcome brake pipe leakage in both the release and service positions of the automatic brake valve.

Propagation: The serial action of a brake application from car to car through a train.

Proper Authority: A position that provides managerial oversight of operation. Proper authority varies by location and type of operation.

Quick Service: The local venting of brake pipe pressure at each car.

Reduction: A decrease of brake pipe pressure at a rate and amount sufficient to initiate or increase an automatic brake application.

Reduction Relay Valve: A device on long cars that helps reduce brake pipe pressure during service and emergency brake applications. The valve compensates for the added length of brake pipes on long cars.

Regulating Valve: At the automatic brake valve, a device that controls the maximum pressure put into the equalizing reservoir and brake pipe. The air controlled is supplied by the locomotive main reservoir and delivered to the automatic brake system.

Run In: The rapid change of the train's coupler slack to a buff state.

Run Out: The rapid change of a train's coupler slack to a draft state.

Running Release: The release of an automatic brake application while the train is moving.

Safety Control Devices: A variety of safety devices including event recorders, alerters, deadman controls, automatic cab signals, cab signal whistles, automatic train stop equipment, and automatic train control equipment.

Sag: A descending grade followed by a level, nearly level, or ascending grade.

Service Brake Application: A reduction of brake pipe pressure at a controlled rate that causes the control valve to move to a service position and apply the brakes.

Slack Action: Movement of part of a coupled train at a speed different from another part of the train.

Solid Block of cars: Two or more freight cars coupled together.

Split Reduction: The preferred method of applying train brakes. An initial 7-psi brake pipe reduction followed by additional reductions in 2 to 3-psi increments spaced 30 seconds apart.

Stretch Braking: Slowing or stopping a train using the automatic air brakes and enough power to maintain a slack-stretched condition not to exceed throttle 4.

Stringlining: The result of excessive draft forces in a train negotiating a curve. Stringlining causes wheel climb, track damage, or derailment.

Throttle Modulation: Adjusting the throttle one notch at a time to control train speed and in-train forces.

Tons per Operative Brake (TOB): The measurement calculated by dividing the gross trailing tonnage of a train, not including locomotives, by the total number of cars with operative air brakes.

Tractive Effort: The force exerted by a locomotive on the track to move a train.

Tractive Effort Meter: The device that indicates the tractive effort being produced by a locomotive with AC traction motors.

Train Consist: Rail cars coupled together which form a train.

Train Separation: Any unplanned train separation involving mechanical coupling equipment.

True Gradient: a term that describes the point, with the automatic brake released, where the pressure on the rear car has stopped rising and the air brake system is fully charged. On distributed power trains, the sum of air flow on the lead and cut in remote consists can be used to indicate when a train has reached true gradient

Undesired Emergency (UDE): An undesired emergency application of a train's air brake system.

Undulating Grade: A track profile with frequent grade changes.

Vent Valve: A valve attached to the brake system of a car or locomotive that will vent brake pipe pressure in response to an emergency brake application.

Wheel Slide: When the wheel rotates slower than lengthwise movement.

Wheel Slip: When the wheel rotates faster than lengthwise movement.

Work Train: A non-revenue service train of 4,000 trailing tons or less used for the administration or upkeep service of the railroad.

Yard Air: A supply of compressed air in a train yard used to charge and test trains.



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